FROM ROBOTS TO COBOTS

Machines · Robots · Collaborative Systems

An educational journey into intelligent automation

OVERVIEW

What We Will Explore

01 **Machines**
Purpose-built devices with predefined operations

02 **Robots**
Complex machines that sense, decide & act

03 **Robot Anatomy**
Sensors · Actuators · Controllers · Structure

04 **Cobots**
Collaborative robots designed to work with people

05 **Key Differences**
How machines, robots and cobots relate

MACHINES

What is a Machine?

- Designed to perform specific, predefined tasks
- Operations are fixed by mechanical, electrical or software design
- Limited or no ability to sense and adapt to environment
- Requires human control or simple programming for each cycle



Example: Washing Machine

EXAMPLE

Washing Machine — A Classic Machine

Purpose

Designed specifically to clean clothes. One primary function.

Control

Controlled by a small embedded program with fixed wash cycles.

Limitation

Cannot sense dirt level or adapt recipe. No environmental awareness.

ROBOTS

What Makes a Robot?

A complex machine capable of sensing its environment, making decisions, and performing a variety of tasks — often with partial or full autonomy.

01	**Sense**	Uses sensors to perceive the world
02	**Decide**	Processes data to choose actions
03	**Act**	Moves and manipulates via actuators
04	**Adapt**	Can learn and update behaviour



APPLICATIONS

Where Robots Excel

Manufacturing

Welding, assembly, material handling, painting

Healthcare

Complex medical surgeries, rehabilitation, logistics

Defence & Security

Surveillance, bomb disposal, reconnaissance

Exploration

Space, underwater, hazardous environments

ANATOMY

Basic Structure of a Robot

01

Sensors

Eyes, ears & touch — gather data from the environment

02

Actuators

Muscles — motors, gears, hydraulics that create motion

03

Processors

Brain — compute decisions from sensor data

04

Outer Body

Skeleton & skin — structure, protection and appearance

HOW IT WORKS

The Sense → Think → Act Loop

SENSE	THINK	ACT	LEARN
Sensors collect data about the environment and the robot itself	Controller processes inputs and decides the next action	Actuators execute the motion or task	Update models / behaviour for better future performance

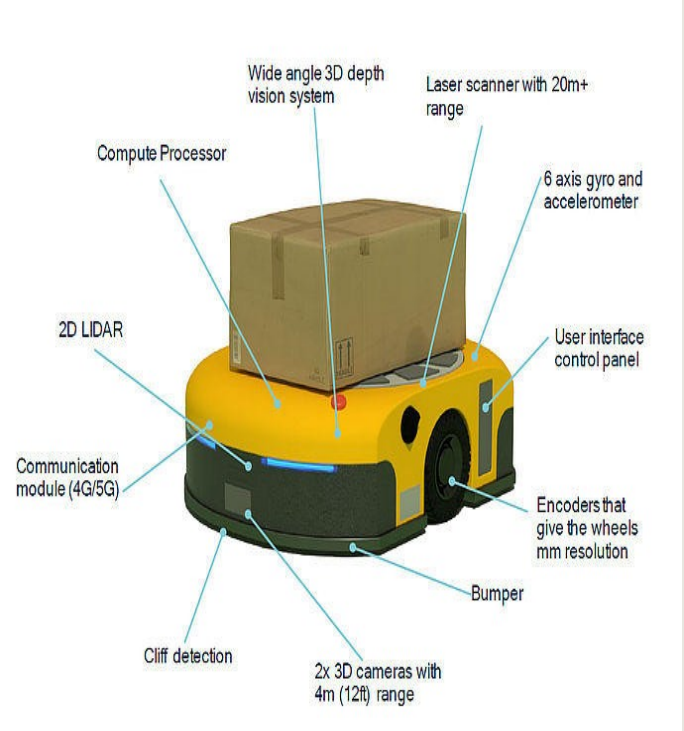
Robots continuously close this loop to interact intelligently with the world.

SENSORS

Sensors — The Robot’s Senses

Enable the robot to learn about its environment, move safely, avoid obstacles and interact with objects.

- Cameras (vision)
- LiDAR / Lasers
- Ultrasonic
- Inertial (IMU)
- Force / Tactile
- Temperature
- Microphones
- Encoders



ACTUATORS

Actuators — Creating Motion

Actuators are the muscles of the robot. They convert electrical signals into physical movement.

Electric Motors

Most common — DC, servo, stepper motors for precise positioning

Gears & Transmissions

Reduce speed, increase torque, change direction of motion

Hydraulics / Pneumatics

High force applications (heavy lifting, industrial presses)

Linear Actuators

Push / pull motion for grippers, lifts and linear joints

CONTROLLERS

Controllers — The Robot’s Brain

- Receives data from all sensors
- Runs algorithms / AI models
- Sends commands to actuators
- Works from pre-written programs or learned policies
- Can be embedded (on-robot) or remote (cloud)

KEY INSIGHT

The controller turns raw sensor numbers into intelligent behaviour.

Without it, a robot is just a pile of motors and cameras.

EXAMPLES

Example: Surveillance Robot Dogs



Mobility

Walks on four legs over complex terrain

Sensing

Cameras, LiDAR, microphones for 360° awareness

Autonomy

Can patrol, follow paths, detect anomalies

Role

Acts as a mobile watchman for security & inspection

COMPARISON

Machines vs Robots

MACHINES

- Single / few fixed tasks
- Predefined operations only
- Little or no sensing
- No decision making
- Human or simple program control
- Example: Washing machine, printer

ROBOTS

- Multiple / complex tasks
- Can adapt behaviour
- Rich sensor suite
- Autonomous decision making
- Sense → Think → Act loop
- Example: Robot arm, Spot dog

COBOTS

Collaborative Robots (Cobots)

Cobots are robots specifically designed to work safely alongside humans, sharing the same workspace and tasks.

Human collaboration	Designed for direct interaction with people
Safety first	Force-limited, soft materials, collision detection
Easy programming	Often teach-by-demonstration, no specialist code
Flexible deployment	Quick to redeploy for different tasks



COMPARISON

Traditional Robots vs Cobots

Aspect	Traditional Robot	Cobot
Workspace	Caged / separated	Shared with humans
Safety	Physical barriers	Inherent (force, speed limits)
Programming	Complex, specialist	Intuitive, teachable
Flexibility	High volume, fixed	High mix, easy re-task
Focus	Full autonomy	Collaboration & assistance

IMPACT

Why Cobots Are Important

Safer workplaces

No cages needed; force sensing prevents injury

Human + Robot teams

Best of both: human creativity + robot precision

Faster ROI

Easier to deploy and reprogram for SMEs

Higher productivity

Humans focus on high-value work, robots on repetitive



SYSTEMS

Multi-Robot Systems

Multiple robots working together (and often with humans) over a network to achieve a common goal.

1

Smart Home / Office

Cleaning robots, security bots and delivery bots coordinating via the cloud

2

Firefighting

Teams of ground and aerial robots mapping fire, delivering water, rescuing people

3

Warehouse / Factory

Fleets of AMRs and cobots sharing maps and tasks in real time

SUMMARY

Key Takeaways

1	**Machines**	Do one (or few) predefined jobs extremely well.
2	**Robots**	Sense, decide and act — they are machines that can adapt.
3	**Anatomy**	Sensors + Actuators + Controllers + Body form the core.
4	**Cobots**	The bridge: robots that safely share space and work with people.
5	**Future**	Multi-robot + human teams will become the new normal.

Thank You

From Machines → Robots → Cobots
The journey of intelligent automation continues

Questions & Discussion

COMPONENTS OF ROBOTS AS A SYSTEM

Focus: Characteristics and Principles of Gears in Robotics

Use of Gears in Robots



Robotic Joints

Allows for smooth and accurate movement. Utilizes spur and harmonic drive gears.



Robot Mobility

Essential for gearbox systems powering robot wheels or tracks for varied terrain.



Lifting Power

Used in mechanisms to magnify force, allowing robots to carry heavy weights.



Accuracy

Ensures precise control and positioning in industrial automation and surgery.

Transmission of Force

The term "force transmission" in robotics describes how mechanical power gets transmitted from one part to another via the interlocking of gear teeth.

When connecting different robotic components, gears are frequently employed to make sure that force and motion generated in one part are effectively transmitted to another.



Amplification of Force

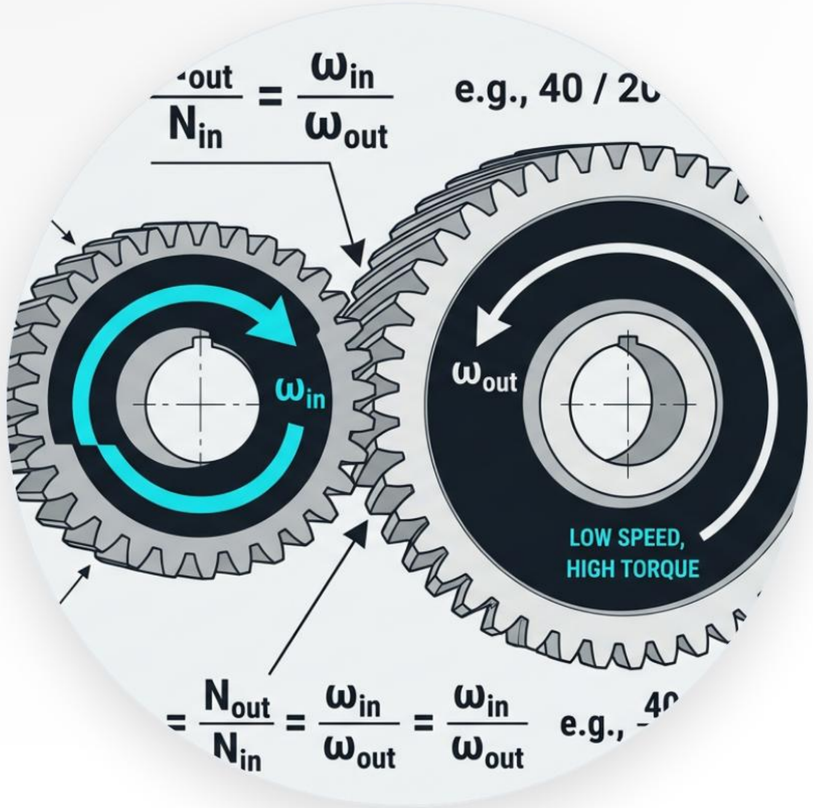


Gears increase force through gear ratios, which differ in the number of teeth on the driver gear (input) and the driven gear (output).

Engineers choose specific gear ratios to:

- Improve **Torque** (rotational force)
- Increase **Rotational Speed**

Principles of Gear Operation



Mechanical Engagement

The primary function of gears is to transmit motion and power from one component or system to another.

Gears are made up of teethed wheels that are a certain size and shape. As these wheels turn, their teeth lock together, moving power from the driver gear to the driven gear.

$$\text{Torque}_{out} = \text{Torque}_{in} \times \frac{N_{driven}}{N_{driver}}$$

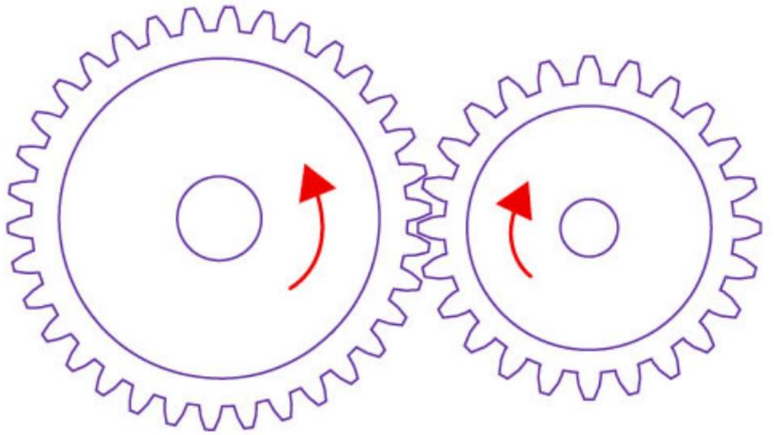
Types of Gears

A classification of common gear designs in robotics

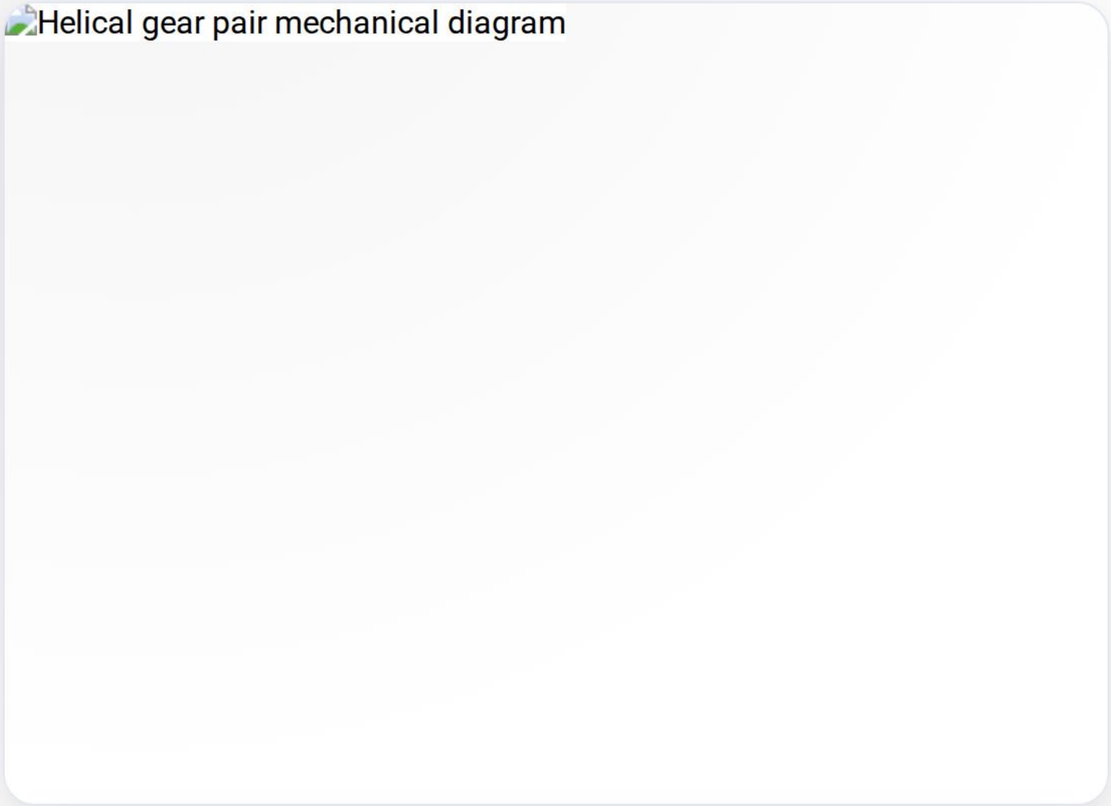
Spur Gears

Spur gears are the most common kind of gears. They have straight teeth and are placed on shafts that are parallel to each other.

- Simplicity in design and manufacturing.
- Highly efficient for power transmission.
- Used widely for general motion in robotics.



Helical Gears



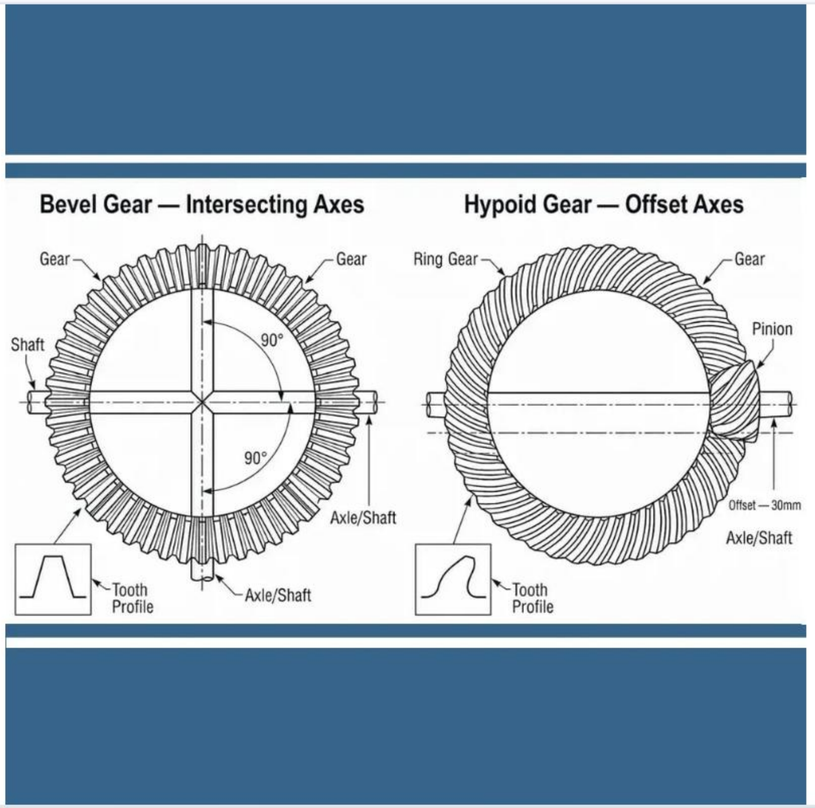
Helical gears are like spur gears, but their teeth are curved. This design choice makes them run more smoothly and quietly.

- Gradual tooth engagement.
- Higher load-carrying capacity.
- Preferred for high-speed robotic applications.

Bevel Gears

Bevel gears are used to alter the direction of movement between two shafts. Their teeth are cut on conical surfaces.

- Allows shafts to intersect at an angle (usually 90°).
- Commonly found in robot steering and differential systems.



Worm Gears



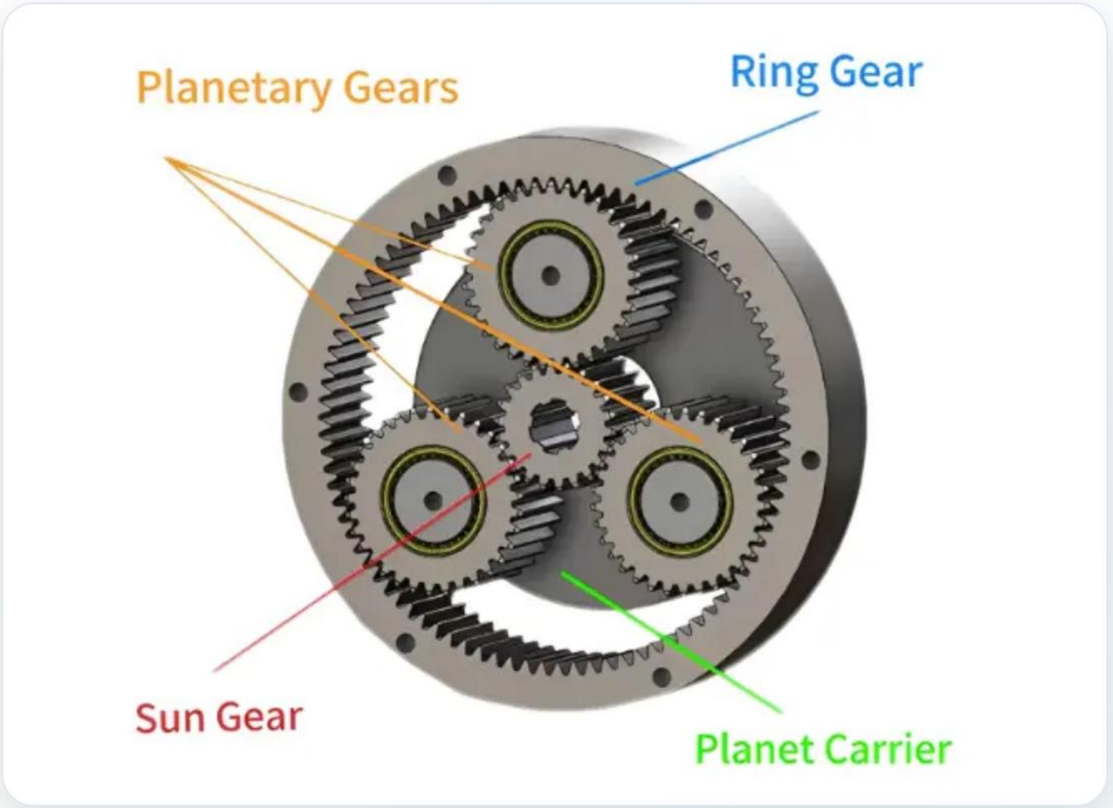
A worm gear is made up of a rod with threads (a "worm") and a wheel with teeth (a "worm gear").

- Enables very high gear reduction ratios.
- Provides "self-locking" features.
- Ideal for applications needing fine positional control.

Planetary Gears

Planetary gears consist of a central "sun" gear, multiple "planet" gears, and an outer "ring" gear.

- Frequently used in robotic arms and small engines.
- Transfer power exceptionally well in a compact space.
- High torque density.



Summary & Questions

Gears are the muscle and precision of robotic systems.

Core Takeaways:

Force Transmission | Torque Amplification | Directional Control

Image Sources



https://static.vecteezy.com/system/resources/thumbnails/070/627/070/small/close-up-of-intricate-interlocking-metal-gears-and-shafts-showcasing-precision-engineering-and-mechanical-complexity-industrial-background-photo.jpeg

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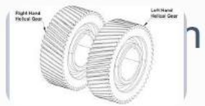
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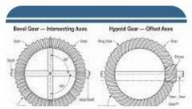
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Image Sources



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MGM`s Polytechnic Aurangabad

COURSE TITAL:-FUNDAMENTALS OF MECHATRONICS

Topic :- Types of Sensor

COURSE CODE:-(11408)

Guided By:-Prof. Kale P.G

MICRO-PROJECT TITAL

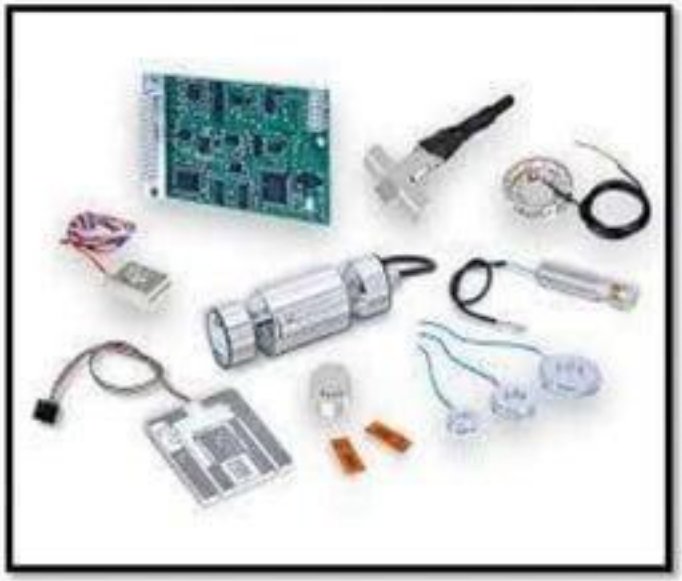
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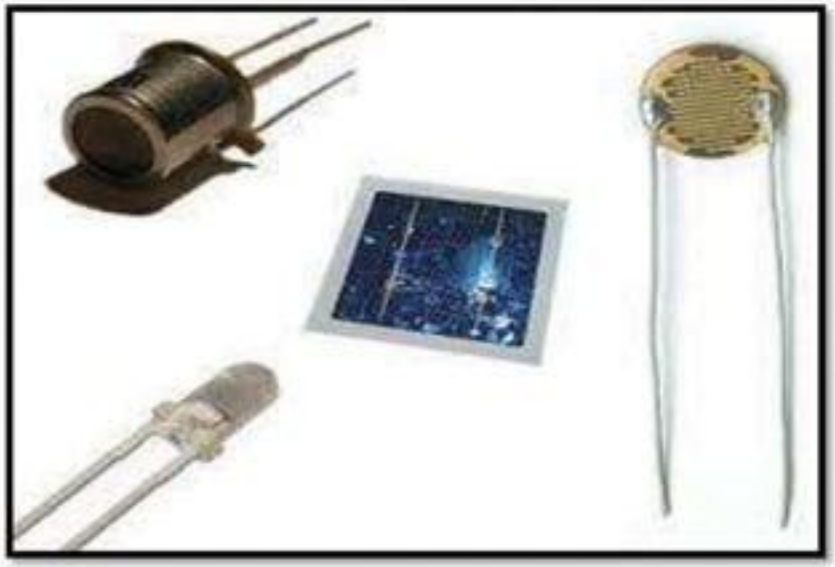
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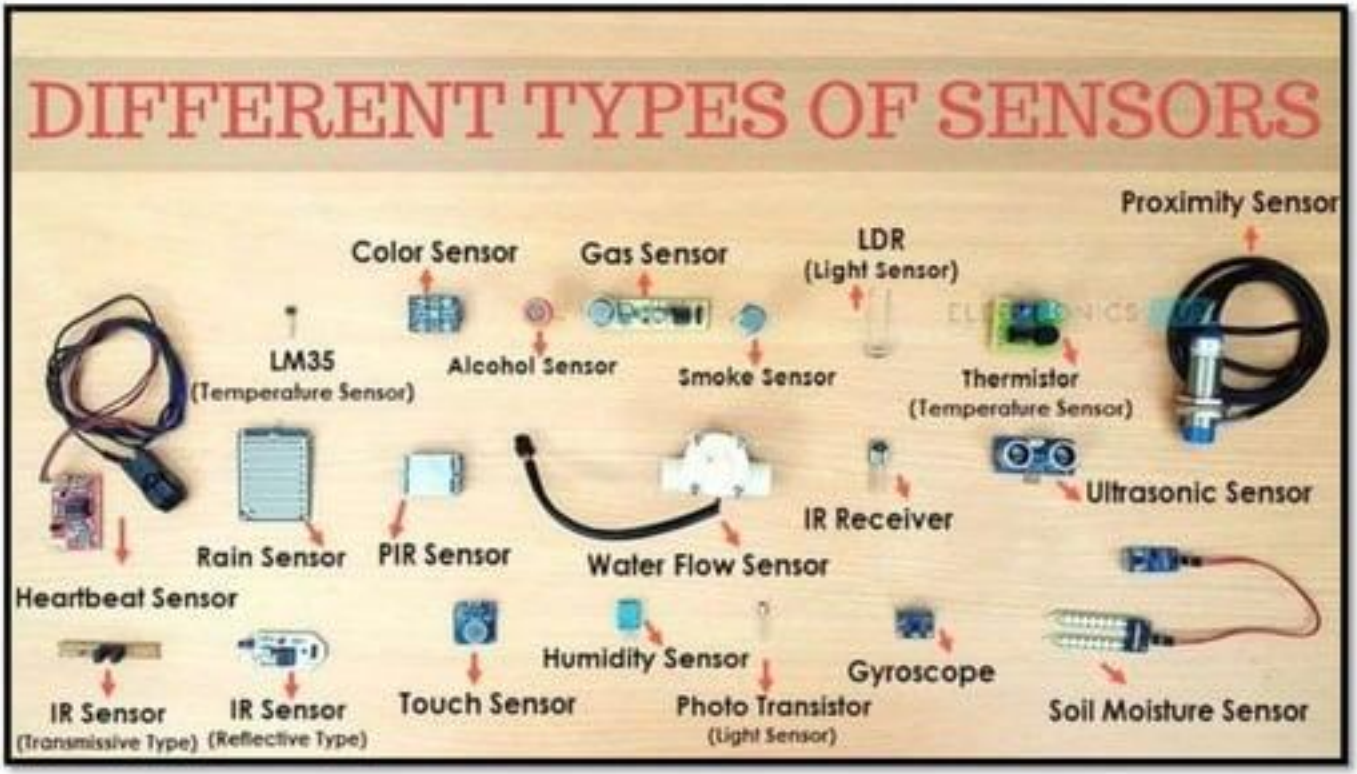
What is the sensor?

A device which detects or measures a physical property and records, indicates, or otherwise responds to it is called sensor





Types of sensors



Types of Sensors Names

- LM35(Temperature sensors)
- Color sensor
- Alcohol sensor
- Smoke sensor
- Ultrasonic sensor
- Soil moisture sensor
- Touch sensor
- Humidity sensor
- Rain sensor
- Proximity sensor
- Gas sensor
- Water flow sensor
- Heartbeat sensors

LM35(Temperature sensors)

- LM35 is a precession Integrated circuit Temperature sensor, whose output voltage varies, based on the temperature around it. It is a small and cheap IC which can be used to measure temperature anywhere between -55°C to 150°C
- Application
- Measuring temperature of a particular environment
- It can be used in HVAC applications as a temperature measurement device.





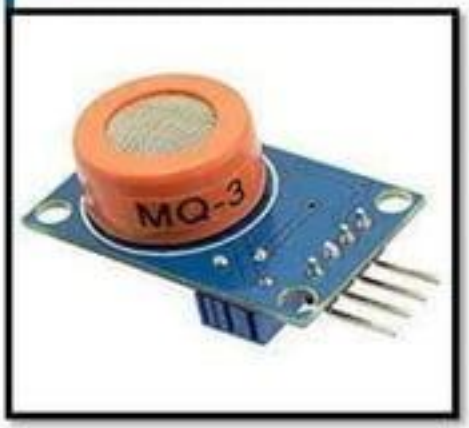
Color sensor

- A color sensor is a type of "photoelectric sensor" which emits light from a transmitter, and then detects the light reflected back from the detection object with a receiver.
- Applications
- Light color temperature measurement
- Medical diagnosis system
- Industrial process control





Alcohol sensor

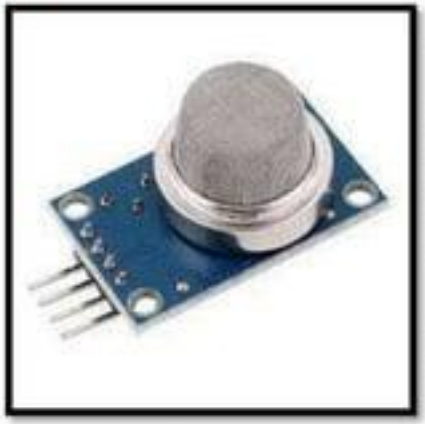


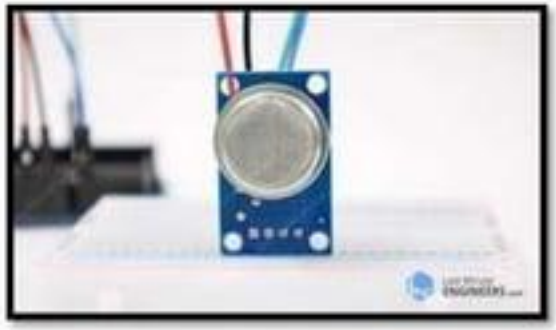
- An alcohol sensor detects the attentiveness of alcohol gas in the air .
- Application
- Alcohol Detector project" can be used in the various vehicles for detecting whether the driver has consumed alcohol or not.
- Use in breath analyzer



Smoke sensor

- A smoke sensor is a device fitted to smoke alarms. A smoke alarm is designed to detect the presence of smoke in a home to alert the occupants that a fire has broken out.
- Application
- They are used in gas leakage detecting equipment in family and industry.
- It is suitable for detecting of LPG ,Natural gas, town gas.
- Avoid the noise of alcohol and cooking fumes cigarette smoke.





Ultrasonic sensor

- An ultrasonic sensor is an electronic device that measures the distance of a target object by emitting ultrasonic sound waves, and converts the reflected sound into an electrical signal.
- Application
- Presence detection.
- People detection for counting.
- Loop control.

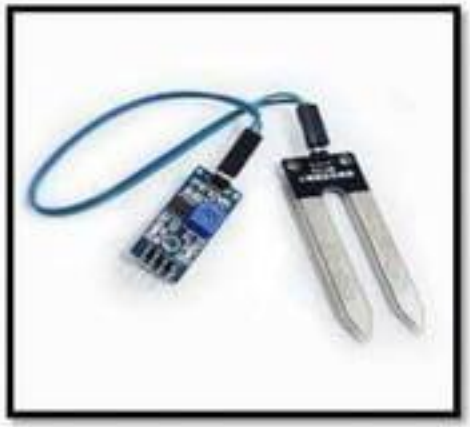




Soil moisture sensor

- The soil moisture sensor is one kind of sensor used to gauge the volumetric content of water within the soil As the straight graven
- Application
- environmental science
- Horticulture
- Biology
- used to measure the volumetric water content of soil





Touch sensor

- A touch sensor is an electronic sensor used in detecting and recording physical touch.

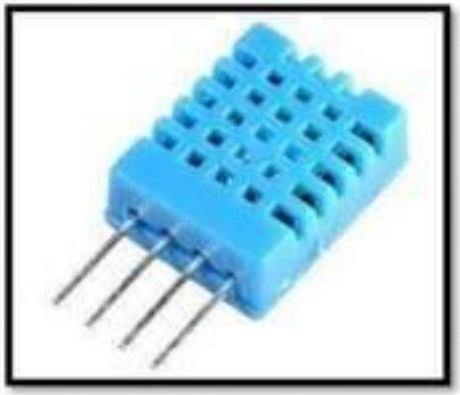


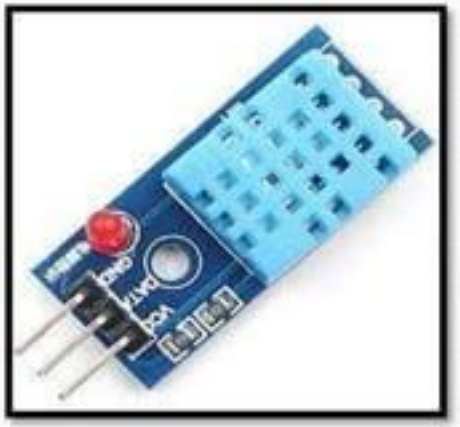
- Application
- Mobile phone.
- monitor screens.
- control panels.



Humidity sensor

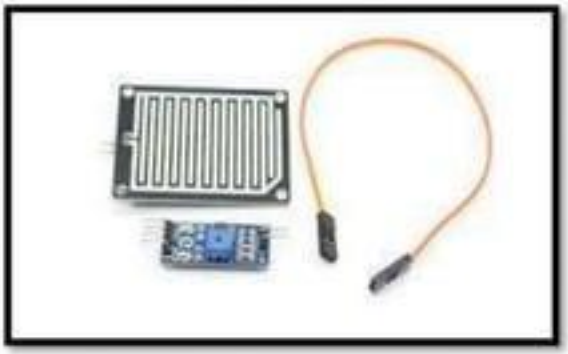
- A humidity sensor is an electronic device that measures the humidity in its environment and converts its findings into a corresponding electrical signal.
- Application
- Climate control for green house.
- Automotive cabin air control
- Food processing





Rain sensor

- A rain sensor is one kind of switching device which is used to detect the rainfall. It works like a switch and the working principle of this sensor is, whenever there is rain, the switch will be normally closed.
- Application
- This sensor is used as a water preservation devices and this is connected to the irrigation system to shut down the system in the event of rainfall.
- This sensor is used to guard the internal parts of an automobile against the rainfall as well as to support the regular windscreen wiper's mode.





Proximity sensor

- A proximity sensor is a device that can detect or sense the approach or presence of nearby objects and for this it does not need physical contact.
- Application
- recycling plants
- self-driving cars
- anti-aircraft systems





Gas sensor

- Gas sensors (also known as gas detectors) are electronic devices that detect and identify different types of gasses. They are commonly used to detect toxic or explosive gasses and measure gas concentration.
- Application
- Process control industries
- Boiler control
- Fire detection
- Alcohol breath tests





Water pressure sensor

- Water pressure sensors are often used to measure the level of water in a tank, or the rate of change in that level.



- Application
- Flow of water
- Flow of Gases
- Flow of speed



Heartbeat sensors

- Heartbeat Sensor is an electronic device that is used to measure the heart rate i.e. speed of the heartbeat.
- Application
- Detect the heart beat
- Heart beat pressure
- Heartbeat speed





Thank you

Functions of Sensors

The "Sensory Organs" of Modern Robotic and Electronic Systems

What is a Sensor?

A device that gathers data from the environment and converts it into a meaningful form interpreted by humans and machines.

Five Primary Functions

Transform

Physical parameters into electrical impulses.

Quantify

Precise measurement of physical values.

Detect

Identify changes or environmental occurrences.

Monitor

Constant reporting on parameter changes over time.

Feedback

Allowing systems to adjust and maintain settings.

1. Physical to Electrical

Sensors act as transducers, converting physical phenomena into signals.

- Temperature to voltage
- Pressure to resistance
- Humidity to frequency
- Position to digital code



2. Precise Quantification

99.9%

Reliability in Precision Data

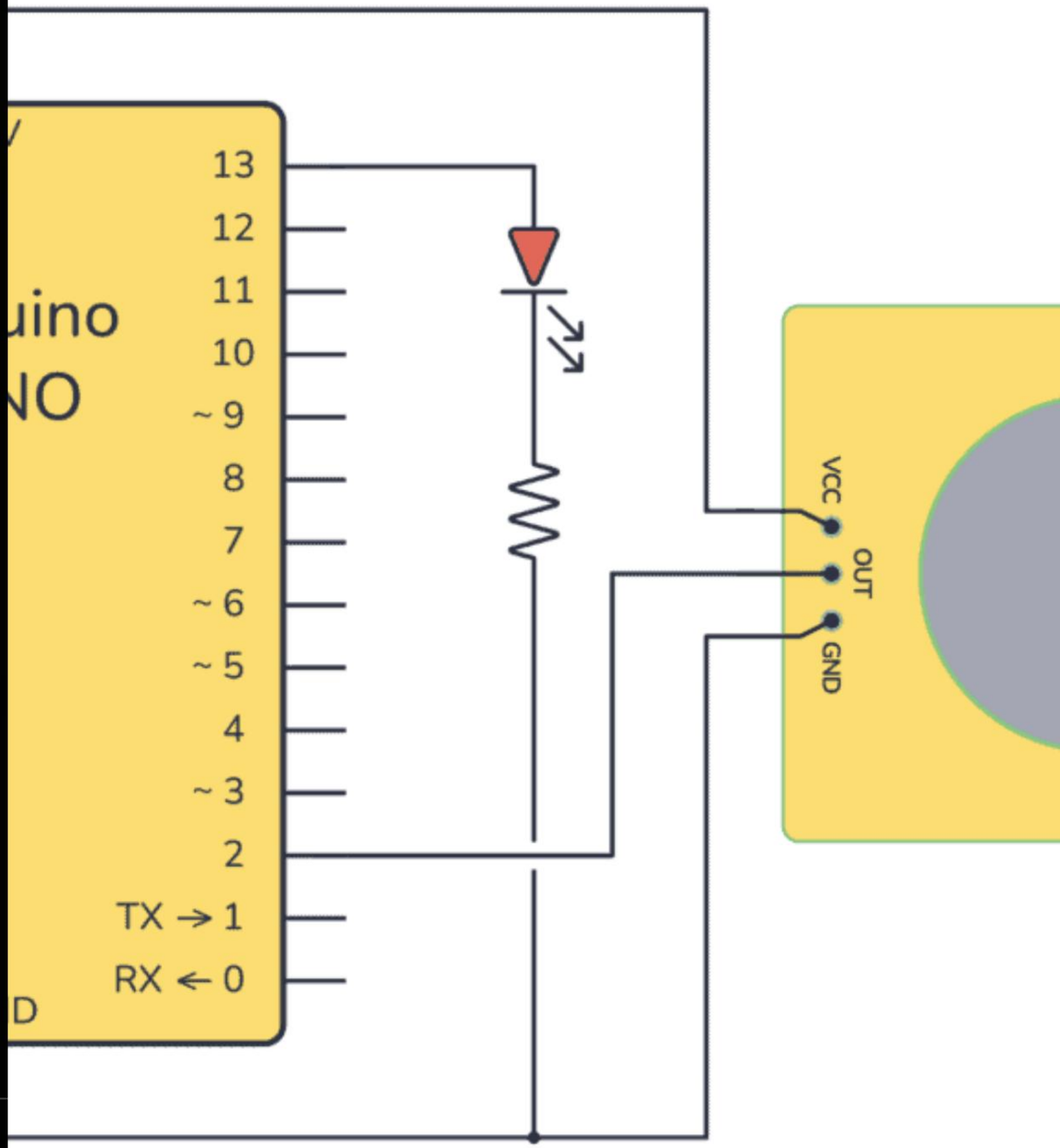
Sensors are designed to perceive specific ranges with high accuracy. They provide the raw data necessary for computation and decision-making.

Without precise quantification, robotic systems would lack the "resolution" needed for fine motor skills and delicate interactions.

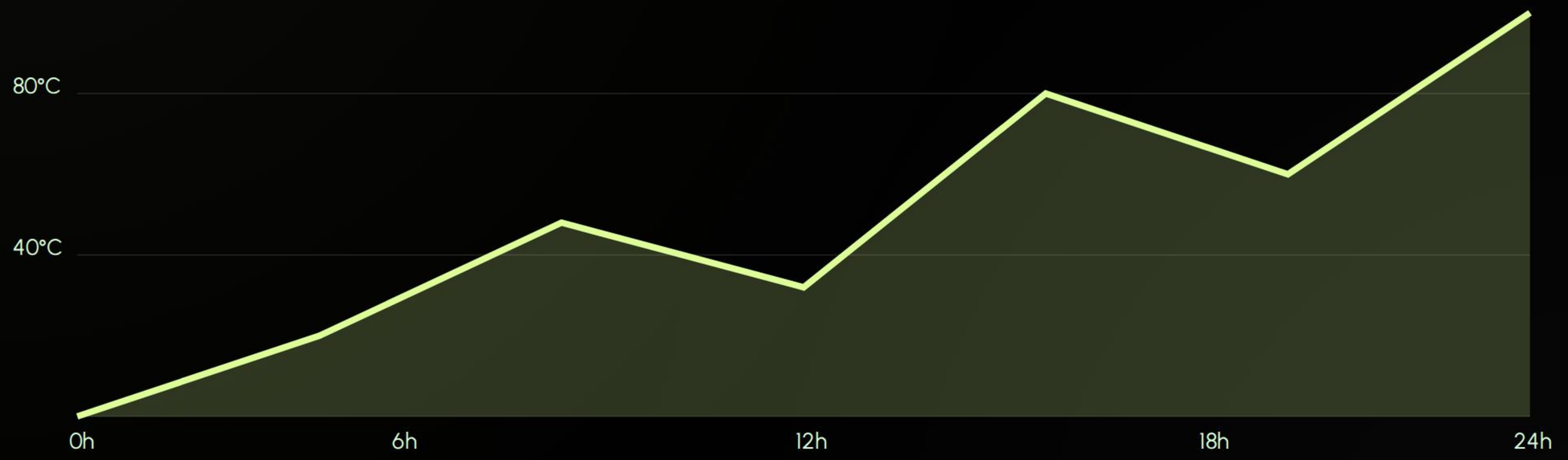
3. Detecting Changes

Sensors excel at recognizing occurrences in the environment, such as the entry of an object or a sudden shift in heat.

This detection triggers electrical signals that act as "interrupts" or events, allowing the controller to react instantly to external stimuli.



4. Constant Monitoring



Sensors report changes over time, creating a historical log of environmental performance.

5. Feedback & Control

Maintain Settings

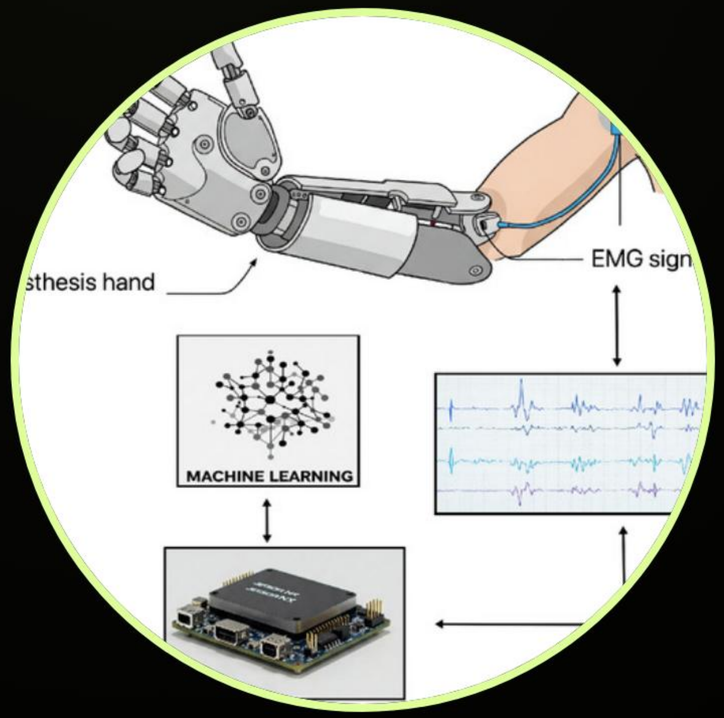
Providing data to ensure the system stays within a designated "Normal" zone (e.g., a thermostat keeping a room at 22°C).

Adjust to Changes

Informing the controller to pivot behavior when external conditions shift (e.g., a robot slowing down when a human is nearby).

Sensors Cross-Domain

Function	Example Sensor	Domain
Transform	Thermocouple	Industrial Heat Control
Quantify	Load Cell	Precision Weighing
Detect	PIR Sensor	Security Systems
Monitor	Hygrometer	Smart Agriculture



The "Senses" of Robotics

Modern robotics depends on a loop of sensing and acting. Without sensors, a robot is "blind" to its own progress and the dangers of its environment.

From tactile skin to LIDAR "eyes", the five functions are integrated into every joint and limb of an advanced robotic system.

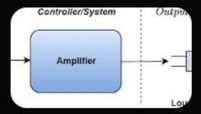
The Sensing Loop

- Reliability: Continuous reporting ensures system health.
- Safety: Immediate detection of occurrences prevents accidents.
- Precision: Quantification allows for micro-level movements.
- Autonomy: Sensors provide the "context" for AI decision making.

Questions ?

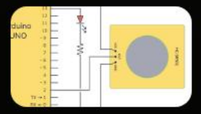
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Image Sources



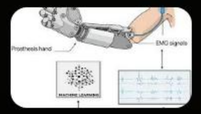
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Human Synergy in Multi-Robot Systems

Unlocking Operational Potential Through Ethical Training, Performance Monitoring, and Adaptive Feedback Loops.

CORE STRATEGIC ADVANTAGES



Ethical Training

Establishing value-based frameworks to enhance robot versatility and mission flexibility.



Real-time Monitoring

Tracking development trajectories and managing resource allocation for peak efficiency.



Adaptive Feedback

Continuous refinement of performance through expert human-in-the-loop oversight.

Section 01

Training for Ethical Goals

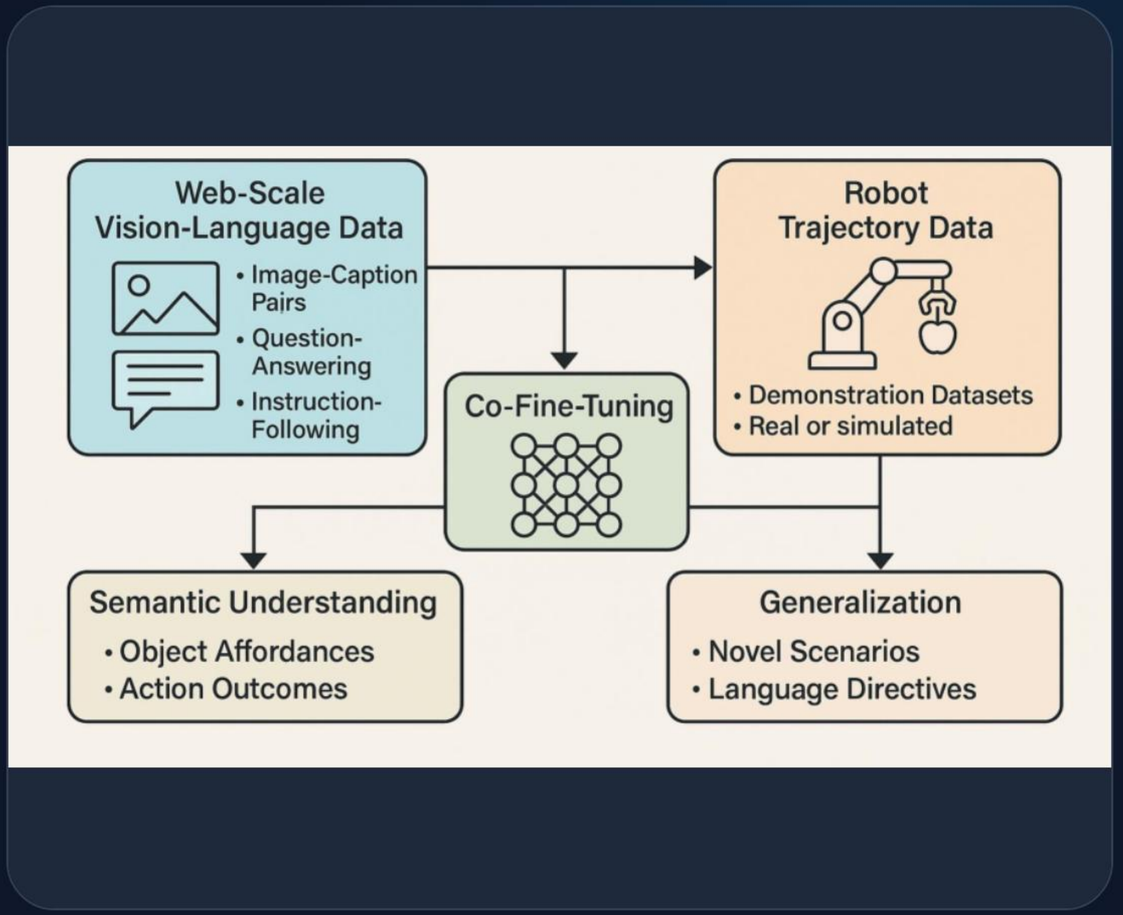
Defining the behavioral constraints and objectives for autonomous multi-agent teams.

ETHICAL GOAL TRAINING

Values-Based Performance

By interacting with human experts, robots can be trained to prioritize ethical goals over raw efficiency.

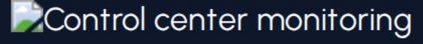
- ✓ Increased versatility in ambiguous environments.
- ✓ Alignment with complex human organizational values.
- ✓ Enhanced flexibility in behavioral adaptation.



Section 02

Supervision & Optimization

Monitoring system development and ensuring effective resource distribution.



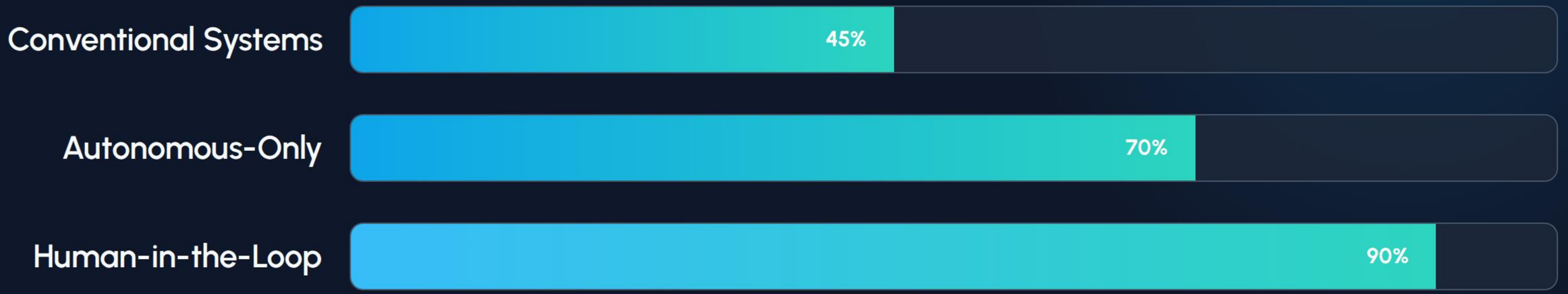
DEVELOPMENT MONITORING

Strategic Oversight

Humans monitor multi-robot development to make informed resource allocation decisions.

This oversight ensures that robots are deployed in the most efficient manner, catching edge cases that autonomous systems might overlook.

EFFICIENCY GAINS VIA HITL



**Comparison of deployment efficiency across different multi-robot management paradigms.*

THE EXPERT PERSPECTIVE



Contextual Judgment

While robots excel at repetition and speed, they lack contextual judgment. Human interaction bridges this gap by providing high-value cognitive oversight.

This is critical in sensitive fields like healthcare, legal processes, and disaster response where ethical accountability is mandatory.

EVOLUTION OF ROBOT LEARNING

Data Phase

Expert demonstration and skill capture.

Training

Ethical goal alignment through interaction.

Deployment

Monitored resource allocation.

Feedback

Real-time performance evaluation.

Optimization

Continuous learning and refinement.

QUANTIFIABLE OUTCOMES

30%

PRODUCTIVITY INCREASE

Efficiency Through Collaboration

Human-robot collaboration (HRC) has been shown to increase productivity by up to 30% in manufacturing and logistics sectors.

- ✓ Reduction in error rates via human verification.
- ✓ Faster adaptation to new tasks.
- ✓ Lower training overhead for complex skills.



The Future of HRI

A future where people and robots collaborate more efficiently, balancing speed and precision with ethics and judgment.

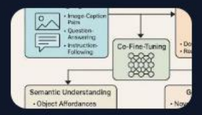
Questions & Discussion

Thank you for your attention.

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IMAGE SOURCES



https://arxiv.org/html/2505.04769v1/x6.png

Source: arxiv.org



https://www.anl.gov/sites/www/files/styles/article_teaser_16x9/public/2024-03/16x9-Hotcell.1.jpg?h=8abcec71&itok=Kjd-tx5_

Source: www.anl.gov



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Source: www.lgt.com



https://static.vecteezy.com/system/resources/previews/066/247/311/non_2x/human-hand-and-robotic-hand-connecting-artificial-intelligence-future-technology-concept-photo.jpg

Source: www.vecteezy.com

Detailed Analysis of Sensors, Actuators and Gears

Components of Robots as a System | Robotics Engineering Fundamentals

Technical Presentation | Detailed Analysis

Sensors • Actuators • Gears • Integration

Agenda

- Introduction to Robot Components as a System
- Part 1: Sensors – Perception, Detection & Feedback
- Vision, Tactile, Temperature, Range, Motion, Navigation, Audio
- Part 2: Actuators – Converting Signals into Motion & Force
- Electric, Hydraulic, Pneumatic & Advanced Actuators
- Part 3: Gears – Power Transmission, Torque & Precision
- Spur, Helical, Bevel, Worm, Planetary, Harmonic Drives
- Integration, Case Studies, Challenges & Future Trends
- Summary, Key Takeaways & Questions

Learning Objectives

- Understand the role of sensors in robotic perception and environmental interaction
- Identify different types of sensors and their working principles (vision, tactile, proximity, inertial, etc.)
- Explain how actuators convert electrical/hydraulic/pneumatic energy into mechanical motion
- Compare actuator types and select appropriate ones based on torque, speed, precision and application
- Describe gear types, gear ratios, and how they amplify torque or change speed/direction
- Analyze the integration of sensors, actuators and gears in complete robotic systems
- Apply knowledge to real-world examples like robotic arms and mobile robots

What are Robot Components?

- Robots are complex mechatronic systems integrating mechanical, electrical, and computational elements
- Core Subsystems:
- Perception: Sensors that gather data about the robot and its environment
- Cognition/Control: Processors, algorithms, and decision-making software
- Action: Actuators and mechanisms that produce movement and force
- Power: Batteries, power supplies, and energy management
- Structure: Chassis, links, joints, and end-effectors
- Transmission: Gears, belts, linkages that transfer power efficiently
- Successful robotics requires harmonious integration of all these components

The Robot Ecosystem: Perception → Decision → Action

Perception Layer

- Sensors collect raw physical data
- Vision systems for object detection
- Proximity & range for obstacle avoidance
- Inertial sensors for orientation & motion
- Tactile & force for physical interaction
- Audio for voice commands & diagnostics

Action Layer

- Actuators generate motion & force
- Motors (DC, servo, stepper) for rotation
- Linear actuators for prismatic motion
- Hydraulic/pneumatic for high power
- Gears multiply torque or speed
- End-effectors (grippers, tools) interact with world

SENSORS

Sensors

Environmental Perception, Detection & Feedback in Robotics

What is a Sensor?

- A sensor is a device that detects and responds to physical stimuli from the environment
- Converts physical quantities (light, sound, pressure, temperature, distance, acceleration) into electrical signals
- Essential for closed-loop control: the robot 'feels' and 'sees' to make intelligent decisions
- Without sensors, robots would be open-loop machines unable to adapt to changing conditions
- Modern robots use dozens to hundreds of sensors for safe and precise operation
- Sensor quality directly impacts robot performance, safety, and autonomy level

Classification of Robotic Sensors

- By Measured Quantity:
  - Vision / Optical: Cameras, IR arrays, LIDAR, photodetectors
  - Proximity & Range: Ultrasonic, IR, capacitive, inductive, magnetic
  - Tactile / Force: Strain gauges, FSRs, load cells, tactile arrays
  - Inertial / Motion: Accelerometers, gyroscopes, IMUs, magnetometers
  - Navigation: GPS, encoders, odometry, SLAM sensors
  - Environmental: Temperature, humidity, gas, pressure
  - Audio: Microphones, microphone arrays, voice activity detectors
- By Output: Analog, Digital, Serial (I2C, SPI, UART), Frequency

Vision Sensors: Introduction

- Vision sensors enable robots to perceive and interpret visual information from surroundings
- Core Capabilities:
- Object Detection: Identifying presence and location of entities in space
- Recognition / Classification: Determining what the object is based on features
- Tracking: Following moving targets over time with continuous updates
- Depth Perception: Stereo vision or structured light for 3D understanding
- Hardware: CMOS/CCD cameras, depth cameras (RGB-D), thermal cameras, event cameras
- Processing: Often paired with onboard AI accelerators for real-time inference (object detection, segmentation)

Vision Sensors: Applications in Robotics

- Industrial: Quality inspection, defect detection, pick-and-place with precise localization
- Autonomous Vehicles & Mobile Robots: Lane detection, traffic sign recognition, pedestrian detection, SLAM
- Human-Robot Interaction: Face recognition, gesture recognition, emotion detection
- Medical & Surgical Robots: Guidance, tissue identification, tool tracking
- Agricultural Robots: Fruit detection & ripeness estimation, weed identification
- Logistics: Package sorting, barcode/QR reading, pallet detection
- Key Challenge: Lighting variations, occlusion, computational cost, real-time requirements

Tactile Sensors

- Tactile sensors mimic the human sense of touch (cutaneous receptors)
- Collect data from physical contact and interaction with objects and environment
- Force / Load Sensors: Measure applied force, weight, or torque (strain gauges, load cells)
- Touch / Contact Sensors: Detect presence of contact, often binary or multi-level (FSRs, capacitive)
- Tactile Arrays / Skins: Dense grids of sensing elements for spatial pressure distribution
- Applications: Gripping force control, slip detection, surface texture recognition, safe human-robot collaboration
- Design inspired by biology for compliant and sensitive robotic hands and skins

Temperature Sensors

- Detect changes in temperature using resistive elements (thermistors, RTDs) or thermocouples
- Critical for protecting robot electronics and maintaining operational limits
- Industrial Applications: Process control, material handling in extreme environments
- Robotic Internal Monitoring: Prevent overheating of motors, drivers, batteries, and processors
- Enclosure Climate Control: Maintain optimal temperature inside robot chassis
- External Sensing: Detect hot/cold objects, thermal mapping (often combined with thermal cameras)
- Common Types: NTC/PTC thermistors, PT100/PT1000 RTDs, K/J thermocouples, digital sensors (DS18B20, TMP36)

Range & Proximity: Ultrasonic Sensors

- Use high-frequency sound waves (typically 40kHz) to detect objects and measure distance
- Non-contact distance measurement – ideal for obstacle avoidance and level sensing
- Working Principle: Emit ultrasonic pulse, measure time-of-flight (TOF) until echo returns
- Distance = (Speed of Sound × Time of Flight) / 2
- Common in parking sensors, mobile robot bumpers, tank level measurement
- Advantages: Works in darkness, dust, and on many surfaces; low cost; simple interface
- Limitations: Affected by soft/absorbent materials, temperature/humidity variations, minimum range

Range & Proximity: IR / Optical Sensors

- Detect presence of objects by emitting and receiving infrared light (or visible)
- Proximity detection when object enters the sensor's field of view and reflects light back
- Applications: Mobile phone proximity (ear detection), home appliances, line following, short-range obstacle avoidance
- Advantages: Fast response, low power, works without physical contact, small form factor
- Types: Reflective (emitter + receiver in one package), Through-beam, Retro-reflective
- Limitations: Affected by ambient light, surface color/reflectivity, dust; shorter range than ultrasonic typically
- Often used alongside ultrasonic for complementary sensing (IR for close range, ultrasonic for longer)

Motion Sensors: Accelerometer

- Measures linear acceleration (including gravity) along one or more axes (typically 3-axis)
- Enables detection of tilt, vibration, shock, and motion intensity
- Smartphone Applications: Auto-rotation of screen, step counting, shake detection, gaming
- Robotics Applications: Balance control (inverted pendulum, walking robots), vibration monitoring, impact detection
- Wearables: Activity tracking, fall detection, gesture recognition
- Key Specifications: Range (±2g to ±16g+), sensitivity, noise density, bandwidth
- Common ICs: MPU6050/9250 (combined with gyro), ADXL345, LIS3DH

Motion Sensors: Gyroscope

- Measures angular velocity (rate of rotation) around one or more axes
- Essential for determining orientation changes and maintaining stability
- Drone Applications: Flight stabilization, attitude control, smooth video gimbal
- VR/AR & Gaming: Head tracking, controller orientation, immersive experiences
- Robotics: Dead-reckoning navigation, balance (self-balancing robots), motion compensation
- MEMS gyroscopes are small, low-cost, and widely integrated into IMUs
- Limitation: Drift over time – usually fused with accelerometer and magnetometer (sensor fusion algorithms like Madgwick, Kalman)

Inertial Measurement Units (IMU)

- Combines accelerometer + gyroscope (often + magnetometer) in a single package
- Provides 6-DoF or 9-DoF motion and orientation data
- Sensor Fusion: Algorithms combine data to compensate for individual sensor weaknesses (drift, noise, gravity)
- Robotics Uses: Odometry, SLAM, balance, trajectory tracking, vibration analysis
- Autonomous Vehicles: INS (Inertial Navigation System) backup when GPS denied
- Popular Modules: MPU6050, MPU9250, BMI160, ICM-20948, BNO055 (with fusion onboard)
- Calibration and proper mounting orientation are critical for accurate results

Navigation Sensors: GPS / GNSS

- Provides global position (latitude, longitude, altitude), velocity, and heading information
- Receives signals from multiple satellites (GPS, GLONASS, Galileo, BeiDou) for trilateration
- Critical for long-range autonomous navigation of vehicles, drones, and outdoor mobile robots
- Typical Accuracy: 2-5 meters (consumer); <1m with RTK/ differential corrections
- Robotics Applications: Waypoint following, geofencing, autonomous delivery, mapping
- Limitations: No indoor/underground operation, multipath errors in urban canyons, power consumption
- Often fused with IMU, wheel odometry, and visual odometry for robust localization

Speech and Audio Processing Sensors

- Microphone Arrays: Multiple mics for beamforming, noise cancellation, and source localization
- Enable voice commands, natural language interaction, and audio event detection
- Voice Activity Detector (VAD): Distinguishes speech from background noise/silence
- Applications: Voice-controlled robots, service robots, meeting assistants, security (glass break, scream detection)
- MEMS Microphones: Small, low-power, digital output (PDM/I2S), widely used in modern devices
- Advanced: Keyword spotting, speaker identification, emotion recognition via AI on edge
- Integration: Often combined with vision for multi-modal human-robot interaction

Velocity and Acceleration Sensing

- Velocity Sensors measure rate of change of position (linear or rotational speed)
- Tachometers: Monitor rotational speed of wheels, motors, shafts (optical, magnetic, hall-effect)
- Applications: Closed-loop motor speed control, odometry for mobile robots, conveyor monitoring
- Doppler Radar / LIDAR: Non-contact speed measurement via frequency shift of reflected waves
- Wheel Encoders: Quadrature encoders provide both speed and direction (ticks per revolution)
- Integration with IMU and GPS improves overall localization and dead-reckoning accuracy
- Important for slip detection, traction control, and precise path following

Additional Important Sensors in Robotics

- Rotary & Linear Encoders: High-precision position feedback for joints and axes
- LIDAR (Light Detection and Ranging): 2D/3D mapping, SLAM, obstacle detection with high accuracy
- Force/Torque Sensors: 6-axis F/T for compliant assembly, grinding, human collaboration
- Current & Voltage Sensors: Motor health monitoring, battery management, power limiting
- Pressure Sensors: Grippers, hydraulics, altitude (barometric), touch
- Chemical / Gas Sensors: Environmental monitoring, leak detection, agriculture
- Event-based Cameras: High-speed, low-power, high dynamic range for fast motion

Core Functions of Sensors

- 1. Transform: Convert physical parameters (temp, pressure, distance, acceleration) into measurable electrical signals
- 2. Quantify: Provide precise, calibrated measurement of the physical quantity
- 3. Detect: Identify presence, absence, or changes in environmental conditions or objects
- 4. Monitor: Continuously report status and variations of the measured parameter over time
- 5. Feedback: Supply real-time data to control systems for closed-loop regulation and adaptation
- These functions enable robots to operate safely, accurately, and autonomously in dynamic environments

Sensor Selection Criteria & Integration

- Selection Factors: Accuracy, resolution, range, response time, power consumption, size, cost, environmental robustness (IP rating, temp range)
- Interface: Analog voltage/current, digital (GPIO, PWM), serial protocols (I2C, SPI, UART, CAN, Ethernet)
- Calibration: Many sensors require initial and periodic calibration for accuracy
- Redundancy: Critical systems use multiple sensors (sensor fusion) for reliability and fault tolerance
- Placement: Mounting location, orientation, and protection from damage/vibration are crucial
- Data Processing: Onboard processing vs raw data streaming; edge AI for vision/audio
- Standards: ROS drivers, IEEE standards, automotive-grade (ISO 26262) for safety-critical applications

ACTUATORS

Actuators

From Electrical, Hydraulic & Pneumatic Signals to Physical Motion & Force

Actuators: From Signal to Motion

- Actuators are the 'muscles' of robots – they convert energy (electrical, hydraulic, pneumatic) into mechanical motion or force
- Essential for achieving movement, manipulation, locomotion, and physical interaction
- Key Performance Metrics: Torque/Force, Speed, Precision/Repeatability, Backlash, Duty cycle, Efficiency, Size/Weight
- Selection depends on application requirements: high speed vs high force, precision positioning vs continuous rotation
- Control: Most electric actuators use PWM, current control, position/velocity feedback loops (PID)
- Safety: Overload protection, thermal protection, compliance (series elastic actuators), emergency stops

Types of Actuators in Robotics

Electric Actuators

- DC Motors (brushed & brushless)
- Servo Motors (position control)
- Stepper Motors (open-loop steps)
- Linear Actuators (lead screw, belt)
- Voice Coil Actuators
- Piezoelectric Actuators
- Advantages: Precise, clean, easy control

Hydraulic & Pneumatic

- Hydraulic: High force, heavy duty
- Pneumatic: Fast, compliant, clean
- Cylinders (linear), rotary actuators
- Require pumps, valves, compressors
- Higher power density
- Used in industrial & heavy robots
- Need fluid maintenance & leaks management

Electric Actuators: DC Motors

- Most common and versatile actuator for robotics – converts electrical energy to rotational mechanical energy
- Brushed DC: Simple, low cost, easy speed control via voltage/PWM; brushes wear over time
- Brushless DC (BLDC): Higher efficiency, longer life, higher speed, requires electronic commutation (ESC)
- Key Parameters: Rated voltage, no-load speed, stall torque, rated torque, power, KV rating (for BLDC)
- Robotics Uses: Wheel drive, joint actuation (with gearbox), fans, pumps, grippers
- Control: H-bridge for direction & PWM for speed; often with encoder for closed-loop position/speed
- Gearboxes are almost always paired to increase torque and reduce speed to useful levels

Electric Actuators: Servo Motors

- Specialized DC motor + gearbox + position feedback sensor (potentiometer or encoder) + control electronics
- Designed for precise angular positioning (typically 0-180° or continuous rotation versions)
- Control Signal: PWM pulse width determines target angle (standard hobby: 1-2ms pulse every 20ms)
- Internal closed-loop control maintains position against load disturbances
- Robotics Applications: Robot arms, pan-tilt mechanisms, walking robots, grippers, camera gimbals
- Advantages: Easy position control without external encoder, holding torque when powered
- Limitations: Limited rotation range (standard), lower speed/torque than industrial servos, quality varies widely

Electric Actuators: Stepper Motors

- Move in discrete steps (typically 1.8° or 0.9° per full step) – excellent for open-loop positioning
- Types: Bipolar (4 wires), Unipolar (5-6 wires); hybrid, permanent magnet, variable reluctance
- Advantages: Precise positioning without feedback (if not overloaded), holding torque at standstill, repeatable
- Disadvantages: Lower efficiency, resonance/vibration issues, torque drops at high speeds, audible noise
- Microstepping: Drivers can divide steps (up to 256x) for smoother, quieter, more precise motion
- Applications: 3D printers, CNC machines, camera sliders, precision linear stages, valve control
- Often paired with encoders for closed-loop stepper systems to detect missed steps

Electric Actuators: Linear Actuators

- Convert rotary motion into linear push/pull motion – essential for prismatic joints and lifting
- Common Mechanisms: Lead screw + nut, ball screw (higher efficiency), belt/pulley, rack & pinion, voice coil
- Integrated versions combine motor, gearbox, screw, and limit switches in one sealed unit
- Applications: Robot arms (prismatic), lifts, hatches, adjustable platforms, medical beds, automation
- Key Specs: Stroke length, force (push/pull), speed, duty cycle, IP rating, feedback (pot or encoder)
- Advantages over rotary + linkage: Direct linear force, simpler design, self-locking (lead screw)
- Voice Coil: Very fast, precise, limited stroke – used in haptics, fast positioning, speakers

Hydraulic Actuators

- Use pressurized hydraulic fluid (oil) to generate very high linear or rotary force/torque
- Highest power density among actuator types – ideal for heavy-duty applications
- Components: Hydraulic pump, reservoir, valves (directional, proportional, servo), cylinders/motors, filters
- Robotics Applications: Large excavators, construction robots, heavy manipulators, flight simulators, presses
- Advantages: Extremely high force, smooth motion, overload protection via relief valves, can hold load
- Challenges: Fluid leaks, maintenance, noise, weight of fluid and components, environmental concerns
- Modern trend: Electro-hydraulic actuators combining electric control with hydraulic power

Pneumatic Actuators

- Use compressed air to produce linear or rotary motion – fast response and inherent compliance
- Common: Pneumatic cylinders (single/double acting), rotary actuators, air muscles, soft grippers
- Advantages: Very fast actuation, clean (no oil), compliant/safe for human interaction, simple, low cost
- Disadvantages: Lower force than hydraulic, compressible air leads to less precise position control, requires compressor
- Applications: Pick-and-place, packaging, soft robotics, collaborative robots (cobots), haptic devices
- Control: Solenoid valves, proportional valves; position control possible with servo-pneumatics + feedback
- Soft Robotics: Pneumatic artificial muscles and inflatable structures enable safe, adaptable robots

Advanced & Emerging Actuators

- Piezoelectric Actuators: Extremely precise (nanometer), fast, but very small stroke; used in optics, micro-positioning
- Shape Memory Alloys (SMA): Contract when heated (Joule effect); used in soft robotics, medical devices, compact mechanisms
- Electroactive Polymers (EAP): Artificial muscles – lightweight, large strain; still emerging for practical robotics
- Dielectric Elastomer Actuators (DEA): High strain, fast, used in soft grippers and haptics
- Series Elastic Actuators (SEA): Add spring compliance for safer interaction and force sensing
- Variable Stiffness Actuators: Change rigidity on demand for different tasks (locomotion vs manipulation)
- Trend: Toward compliant, backdrivable, and sensor-rich actuators for safe human-robot collaboration

Actuator Selection Criteria

- Required Force/Torque and Speed profile (continuous vs peak)
- Precision & Repeatability needs (positioning accuracy, backlash tolerance)
- Duty Cycle and Thermal Management (continuous operation vs intermittent)
- Power Source & Efficiency (battery life critical for mobile robots)
- Size, Weight, and Mounting Constraints
- Environmental Conditions (temperature, dust, moisture, chemicals, radiation)
- Cost, Availability, Maintenance Requirements, and Integration Complexity (drivers, gearboxes, feedback)
- Safety: Backdrivability, compliance, fail-safe behavior, certifications

GEARS & TRANSMISSION

Gears & Transmission

Amplifying Torque, Changing Speed & Direction, Enabling Precision Motion

Why Gears Matter in Robotics

- Gears are fundamental mechanical components that transmit power and motion between rotating shafts
- Primary Functions:
- Force/Torque Amplification: Increase output torque at expense of speed (or vice versa)
- Speed Reduction/Increase: Match motor characteristics to load requirements
- Direction Change: Bevel gears, worm gears change axis of rotation
- Precision Positioning: Low backlash gears enable accurate joint control
- Without proper gearing, most motors cannot deliver usable torque/speed for robotic joints and wheels
- Gear choice dramatically affects robot performance, efficiency, noise, and cost

Use of Gears in Robots

- Robotic Joints: Provide smooth, accurate, high-torque motion at each degree of freedom (harmonic, planetary, cycloidal)
- Robot Mobility: Gearboxes in wheel drives and tracks for torque multiplication on varied terrain
- Lifting & Manipulation: Magnify force so robots can lift heavy payloads or apply strong gripping force
- Accuracy & Repeatability: Precision gears (low backlash) critical for industrial robots and surgical systems
- Power Distribution: Multiple output shafts from single motor via gear trains
- Compliance & Safety: Some gearboxes designed with elasticity or backdrivability for safe interaction

Transmission of Force through Gears

- Gears transmit mechanical power via interlocking teeth – force is transferred at the point of contact
- Driver (input) gear imparts motion and force to the driven (output) gear through tooth engagement
- Power is conserved (ideally): Power_in ≈ Power_out (minus efficiency losses from friction, heat)
- Torque and angular velocity are inversely related: higher torque output means lower speed output
- Proper tooth profile (involute is standard) ensures smooth, constant velocity ratio and minimal wear
- Lubrication, material (steel, plastic, brass), and manufacturing quality (AGMA/DIN standards) affect efficiency and life

Amplification of Force: Gear Ratio

- Gear Ratio (GR) = Number of teeth on driven gear / Number of teeth on driver gear
- Torque_out = Torque_in × GR × Efficiency
- Speed_out = Speed_in / GR
- Engineers select GR to optimize for the application:
- High GR: High torque, low speed – lifting, heavy manipulation, precise low-speed positioning
- Low GR: High speed, lower torque – fast motion, wheels for mobile robots
- Compound gear trains (multiple stages) achieve very high ratios in compact space
- Example: 10:1 first stage + 5:1 second stage = overall 50:1 ratio

Gear Ratio & Torque Relationship

$$\text{Torque}_{out} = \text{Torque}_{in} \times (N_{driven} / N_{driver}) \times \eta$$

Where N = number of teeth, η = efficiency (typically 0.9-0.98 per stage). Speed is inversely proportional. This fundamental relationship guides all gearbox design in robotics – choosing the right ratio balances torque, speed, and motor sizing.

Principles of Gear Operation

- Mechanical Engagement: Teeth of driver and driven gears mesh, transferring motion and power without slippage (positive drive)
- Constant Velocity Ratio: Properly designed gears maintain fixed speed ratio throughout rotation (involute profile)
- Pressure Angle: Standard 20° or 25° – affects force direction and tooth strength
- Backlash: Small clearance between teeth to prevent binding; too much reduces precision and causes noise/impact
- Efficiency: 90-98% per stage depending on type, lubrication, load, and manufacturing quality
- Key Design Considerations: Tooth strength (bending, pitting), contact ratio, noise, thermal, lubrication

Types of Gears: Overview

- Spur Gears: Straight teeth, parallel shafts – simplest, most common, efficient
- Helical Gears: Angled teeth – smoother, quieter, higher load capacity, axial thrust
- Bevel Gears: Conical shape – change direction of shaft axis (usually 90°)
- Worm Gears: Worm (screw) + wheel – very high reduction, self-locking, right-angle drive
- Planetary (Epicyclic) Gears: Sun + planet + ring – compact, high torque density, multiple ratios possible
- Harmonic Drive: Strain wave gearing – zero backlash, high precision, compact, widely used in robot joints
- Other: Cycloidal, hypoid, rack & pinion, internal gears, face gears

Spur Gears

- Most common and simplest gear type – straight teeth parallel to the axis of rotation
- Shafts must be parallel; teeth mesh directly
- Advantages: Simple design & manufacturing, high efficiency (up to 98%), no axial thrust, low cost
- Disadvantages: Noisy at high speeds (sudden tooth engagement), lower load capacity than helical
- Robotics Applications: General purpose gearboxes, prototypes, educational robots, low-speed high-torque joints
- Variants: External, internal (ring), rack & pinion (converts rotary to linear)
- Quality grades (AGMA) determine precision, noise, and life – higher grades for precision robotics

Helical Gears

- Teeth are cut at an angle (helix) to the axis – gradual tooth engagement
- Advantages over spur: Smoother & quieter operation, higher load-carrying capacity, better tooth strength
- Disadvantages: Axial thrust force requires thrust bearings; slightly lower efficiency; more complex manufacturing
- Double-helical (herringbone): Cancels axial thrust – used in high-power applications
- Robotics Applications: High-speed robot joints, precision gearboxes, automotive transmissions, industrial machinery
- Preferred when noise reduction and smooth motion are priorities (e.g., collaborative robots, medical devices)

Bevel Gears

- Teeth cut on conical surfaces – designed to transmit motion between intersecting shafts (usually 90°)
- Straight bevel: Simplest, similar to spur but conical
- Spiral bevel: Curved teeth like helical – smoother, quieter, higher capacity (most common in vehicles)
- Hypoid: Offset axes (not intersecting) – allows lower driveshaft in vehicles, higher strength
- Robotics Applications: Robot steering differentials, right-angle drives, wrist joints, camera pan-tilt
- Advantages: Compact right-angle power transmission, good efficiency
- Considerations: Requires precise alignment and proper lubrication; thrust forces present

Worm Gears

- Consist of a worm (threaded screw-like shaft) meshing with a worm wheel (gear)
- Very high gear reduction ratios possible in single stage (5:1 to 100:1+)
- Self-locking: Often cannot be back-driven (worm drives wheel but wheel cannot easily drive worm) – great for holding position without brakes
- Advantages: High reduction in compact space, self-locking, quiet, shock absorption
- Disadvantages: Lower efficiency (sliding contact), heat generation, requires good lubrication, unidirectional often
- Robotics Applications: Heavy lifting mechanisms, winches, positioners where holding torque without power is needed, tuning mechanisms

Planetary (Epicyclic) Gears

- Consist of central sun gear, multiple planet gears on a carrier, and outer ring gear (annulus)
- Multiple configurations: Sun input + ring fixed = high reduction; ring input + sun fixed = overdrive, etc.
- Advantages: Very compact for high torque, high torque density, coaxial input/output, multiple ratio options, good efficiency
- Disadvantages: More complex, higher manufacturing precision needed, can be noisy if not well made
- Robotics Applications: Robot arm joints, wheel drives, winches, high-torque compact gearboxes, automatic transmissions
- Widely used in industrial robots and precision positioning systems for excellent power-to-size ratio

Harmonic Drive (Strain Wave) Gears

- Special high-precision gearing using elastic deformation of a flexspline inside a circular spline with elliptical wave generator
- Key Features: Near-zero backlash, high torque capacity in small package, high precision/repeatability, excellent efficiency for precision type
- Typical Ratios: 30:1 to 160:1 or higher in single stage – very compact
- Robotics Applications: Almost standard in high-end robot arms, collaborative robots, precision positioning, aerospace, medical robots
- Advantages: Zero backlash (critical for precision), high torsional stiffness, compact & lightweight, long life
- Considerations: Higher cost, lower torque capacity than some planetary at same size, temperature sensitivity of flexspline

Gear Selection & Design Considerations

- Torque & Speed Requirements: Calculate required output torque and speed; size motor + gearbox accordingly
- Precision Needs: Backlash tolerance – zero backlash (harmonic) for precision assembly/surgery; higher backlash ok for mobile robots
- Efficiency: Important for battery-powered robots; helical/planetary better than worm for continuous duty
- Size, Weight, Cost: Planetary and harmonic offer best torque/volume; worm for high ratio low cost
- Duty Cycle & Thermal: Continuous high power needs cooling/lubrication consideration
- Environment: IP rating, temperature range, corrosion, shock/vibration resistance
- Backdrivability: Needed for compliant/safe robots; affects choice (planetary more backdrivable than worm)

Gearboxes & Integrated Drives in Robotic Systems

- Most robotic joints use integrated motor + gearbox units (servo actuators) for compactness and performance
- Common in industrial arms: Harmonic or planetary + servo motor + encoder + brake + controller in one housing
- Mobile robots: Wheel hub motors with integrated planetary or cycloidal gearboxes
- Trends: Hollow-shaft actuators (cables pass through center), integrated torque sensors, series elastic designs
- Maintenance: Proper lubrication, sealing, and alignment critical for long life and performance
- Standards: Many robot manufacturers use proprietary or standardized actuator modules (e.g., from Harmonic Drive, Maxon, Kollmorgen)

Case Study: Robotic Arm Component Integration

- Joints: Harmonic drive or high-precision planetary gearboxes + servo/BLDC motors for zero-backlash positioning
- Sensors: Absolute encoders or resolvers on each joint for precise angle feedback; torque sensors for compliant control
- End-effector: Force/torque sensor + tactile sensors + vision (wrist camera) for grasping and insertion tasks
- Control: High-frequency PID or advanced (impedance, admittance) control loops using sensor feedback
- Integration Challenge: Cable management through hollow actuators, thermal management, vibration isolation
- Result: Sub-millimeter repeatability, safe human collaboration (if cobot), high payload-to-weight ratio

Case Study: Mobile Robot Drivetrain & Perception

- Drive: DC or BLDC motors + encoders + planetary gearboxes for each wheel (differential drive common)
- Sensors: LIDAR + cameras + IMU + ultrasonic/IR for mapping, localization, and obstacle avoidance
- Navigation: GPS (outdoor) or visual/ laser SLAM (indoor) fused with wheel odometry and IMU
- Control: ROS or custom stack – sensor fusion (EKF), path planning, velocity commands to motor controllers
- Power: Battery + BMS with current/voltage monitoring; efficient gearboxes extend runtime
- Safety: Bump sensors, emergency stop, cliff detection, speed limiting based on sensor confidence

Challenges in Component Design & Integration

- Sensor Noise & Drift: Requires filtering, calibration, and sensor fusion algorithms
- Actuator Limitations: Backlash, compliance, thermal limits, saturation – must be modeled in control
- Gear Issues: Backlash, wear, efficiency losses, noise, thermal expansion affect precision over time
- Integration Complexity: Wiring, communication protocols, real-time requirements, EMI
- Power & Thermal: High-performance actuators generate heat; sensors and electronics need stable conditions
- Cost vs Performance: High-precision components (harmonic, LIDAR, torque sensors) are expensive
- Reliability & Safety: Redundancy, diagnostics, fail-safes, standards compliance (ISO 10218, ISO/TS 15066 for cobots)

Emerging Trends in Robot Components

- Smart Sensors: Onboard processing, AI edge inference, self-calibration, digital twins
- Soft & Compliant Actuators: Pneumatic muscles, DEAs, SMAs for safer human interaction and adaptability
- Additive Manufacturing: 3D printed custom gears, lightweight structures, integrated sensors
- Miniaturization: MEMS, chip-scale actuators and sensors enabling micro-robots and wearables
- Wireless & Energy Harvesting: Reducing wiring, self-powered sensors
- Digitalization: Digital twins, predictive maintenance using sensor data + ML
- Sustainability: Recyclable materials, efficient designs, biodegradable soft robots

Summary: Key Takeaways

Sensors

Enable perception & closed-loop control. Choose type based on measured quantity, accuracy, environment.

Actuators

Convert energy to motion. Match type (electric/hydraulic/pneumatic) to force, speed, precision needs.

Gears

Amplify torque, change speed/direction. Select for backlash, efficiency, compactness, self-locking.

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Core Principles Recap

- Sensors: Transform physical world into data → Quantify → Detect changes → Monitor → Provide feedback for control
- Actuators: Accept control signals → Convert energy form → Produce controlled motion or force → Enable action
- Gears: Transmit power mechanically → Multiply torque or speed → Change direction → Enable precision & efficiency
- System View: Robot performance = f(Sensor quality × Actuator capability × Transmission efficiency × Control intelligence)
- Design is always a trade-off: cost vs performance, size vs power, precision vs robustness, speed vs torque

Questions?

Thank you for attending this detailed overview of robot components and systems integration.

Source: Components of Robots as a System | Robotics Unit
Expanded Technical Presentation - Sensors, Actuators & Gears

References & Further Reading

- Robotics: Modelling, Planning and Control (Siciliano et al.) – Standard textbook
- Introduction to Robotics: Mechanics and Control (Craig) – Classic reference
- Springer Handbook of Robotics – Comprehensive coverage
- Manufacturer resources: Harmonic Drive, Maxon Motor, Kollmorgen, Sick, Cognex, Intel RealSense
- ROS (Robot Operating System) documentation and tutorials – Practical integration
- IEEE Robotics and Automation Society publications & conferences (ICRA, IROS, CASE)
- Standards: ISO 10218 (Industrial robots), ISO/TS 15066 (Collaborative robots)

Sensor Comparison Table (Summary)

- Vision: High information density, computationally heavy, lighting dependent – Best for recognition & mapping
- Ultrasonic/IR: Low cost, simple, fast – Best for proximity/obstacle avoidance
- Tactile/Force: Direct physical interaction data – Essential for manipulation & safety
- IMU: High bandwidth motion data, drift issues – Core for balance, odometry, stabilization
- GPS: Global absolute position (outdoors) – Critical for long-range autonomous navigation
- Encoders: Precise joint/wheel position & velocity – Fundamental for control loops
- LIDAR: Accurate 2D/3D ranging & mapping – Gold standard for mobile robot perception

Actuator Comparison Quick Reference

- DC Motor + Gearbox: Versatile, cost-effective, needs encoder for position – General purpose
- Servo: Easy position control, holding torque, limited range – Arms, pan-tilt, small robots
- Stepper: Open-loop precise steps, resonance issues – 3D printers, precision stages
- BLDC + ESC: High efficiency, high speed, needs commutation – Drones, high-performance drives
- Hydraulic: Highest force density – Heavy machinery, large robots
- Pneumatic: Fast, compliant, clean – Packaging, soft robotics, cobots
- Piezo/SMA: Ultra-precise or compact – Micro-robotics, optics, medical devices

Gear Type Quick Selection Guide

- Spur: Simple, efficient, parallel shafts, moderate precision – Prototypes, general use
- Helical: Smooth, quiet, high load – High-speed or noise-sensitive applications
- Bevel: Right-angle drive – Steering, wrists, compact direction change
- Worm: High ratio, self-locking, right-angle – Lifting, holding position without power
- Planetary: Compact high torque, coaxial – Robot joints, wheel drives, winches
- Harmonic: Zero backlash, high precision, compact – Precision arms, medical, aerospace
- Rule of thumb: Start with planetary or harmonic for most robotic joint applications

Practical Design Workflow

- 1. Define task requirements: workspace, payload, speed, precision, environment, duty cycle, safety
- 2. Select sensors: What data is needed? Accuracy? Update rate? Cost/power constraints?
- 3. Size actuators: Calculate required torque/speed at each joint/drive; include safety factor
- 4. Choose transmission: Gear type/ratio to match motor to load; consider backlash & efficiency
- 5. Control architecture: Feedback loops, sensor fusion, real-time OS or ROS, safety monitors
- 6. Integration & testing: Mechanical assembly, wiring, calibration, tuning, validation
- 7. Iterate: Real-world performance often reveals needed adjustments in gearing, sensor placement, or control gains

Common Pitfalls to Avoid

- Undersizing actuators or gearboxes (leading to overheating, missed steps, or failure under load)
- Ignoring backlash in precision applications (causes positioning errors and vibration)
- Poor sensor placement or calibration (garbage data → poor control)
- Insufficient cooling or duty cycle consideration for high-power actuators
- Overly complex sensor fusion without proper tuning (can be worse than single sensor)
- Neglecting cable management and strain relief in moving joints
- Choosing components based only on datasheet specs without real-world testing

Resources for Further Learning

- Online: Coursera/edX robotics courses, MIT OpenCourseWare, Stanford CS223A
- YouTube: Channels on robot building, mechanism design, ROS tutorials
- Communities: Reddit r/robotics, ROS Discourse, RobotShop forum, Hackaday
- Books: 'Robotics' by Bruno Siciliano, 'Modern Robotics' by Lynch & Park (free PDF available)
- Tools: Onshape/Fusion 360 for CAD, Gazebo/Isaac Sim for simulation, Arduino/Raspberry Pi for prototyping
- Kits: LEGO Mindstorms, VEX, TurtleBot, Franka Emika for hands-on experience
- Professional: Attend ICRA, IROS, or local robotics meetups and workshops

Final Checklist for Robot Component Design

- ■ Sensors cover all necessary perception modalities with adequate accuracy and update rate
- ■ Actuators sized with margin for payload, acceleration, and contingencies
- ■ Gear ratios optimized for torque-speed curve of chosen motors
- ■ Backlash, compliance, and efficiency considered for the application precision needs
- ■ Power budget, thermal management, and duty cycle validated
- ■ Safety features (E-stop, torque limiting, collision detection) implemented
- ■ Calibration procedures, diagnostics, and maintenance plan defined
- ■ Full system tested in realistic conditions before deployment

Appendix: Common Sensor Interfaces

- Analog: Voltage output (0-5V, 0-10V) or current loop (4-20mA) – simple but noise susceptible
- Digital GPIO: On/Off, PWM, frequency, pulse counting – proximity, limit switches, tachometers
- I2C: Multi-device bus, easy wiring – IMUs, temperature, many MEMS sensors
- SPI: High speed, short distance – high-performance IMUs, ADCs, flash memory
- UART/Serial: Simple point-to-point – GPS, some LIDARs, debug consoles
- CAN Bus: Robust automotive/industrial – motor controllers, distributed sensors
- USB/Ethernet: High bandwidth – cameras, 3D sensors, complex devices

Appendix: Gear Terminology Quick Reference

- Pitch Circle: Imaginary circle where teeth mesh; defines pitch diameter
- Module (m): Pitch diameter / number of teeth (metric); inversely related to tooth size
- Pressure Angle: Angle between line of centers and line of action (standard 20°)
- Addendum / Dedendum: Height of tooth above/below pitch circle
- Backlash: Clearance between mating teeth (intentional for lubrication & thermal expansion)
- Contact Ratio: Average number of teeth in contact – higher = smoother operation
- AGMA / DIN Quality: Gear precision grade; higher number = tighter tolerances, quieter, more expensive

Questions?

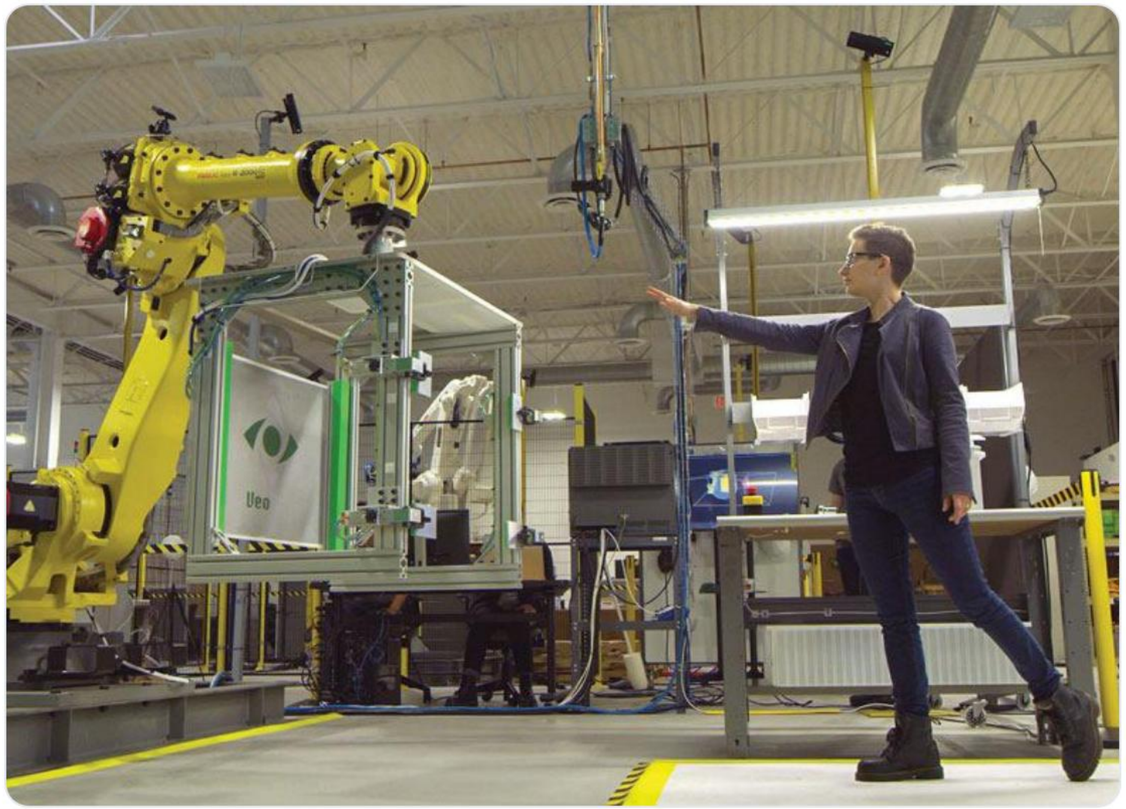
Thank you for attending this detailed overview of robot components and systems integration.

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Expanded Technical Presentation - Sensors, Actuators & Gears

Importance of Collaborative Robots (Cobots)

1. Enhanced Safety

- Designed to work alongside humans.
- Built-in safety mechanisms prevent injury.
- Safe upon direct contact.
- Safer alternative to standard industrial robots.



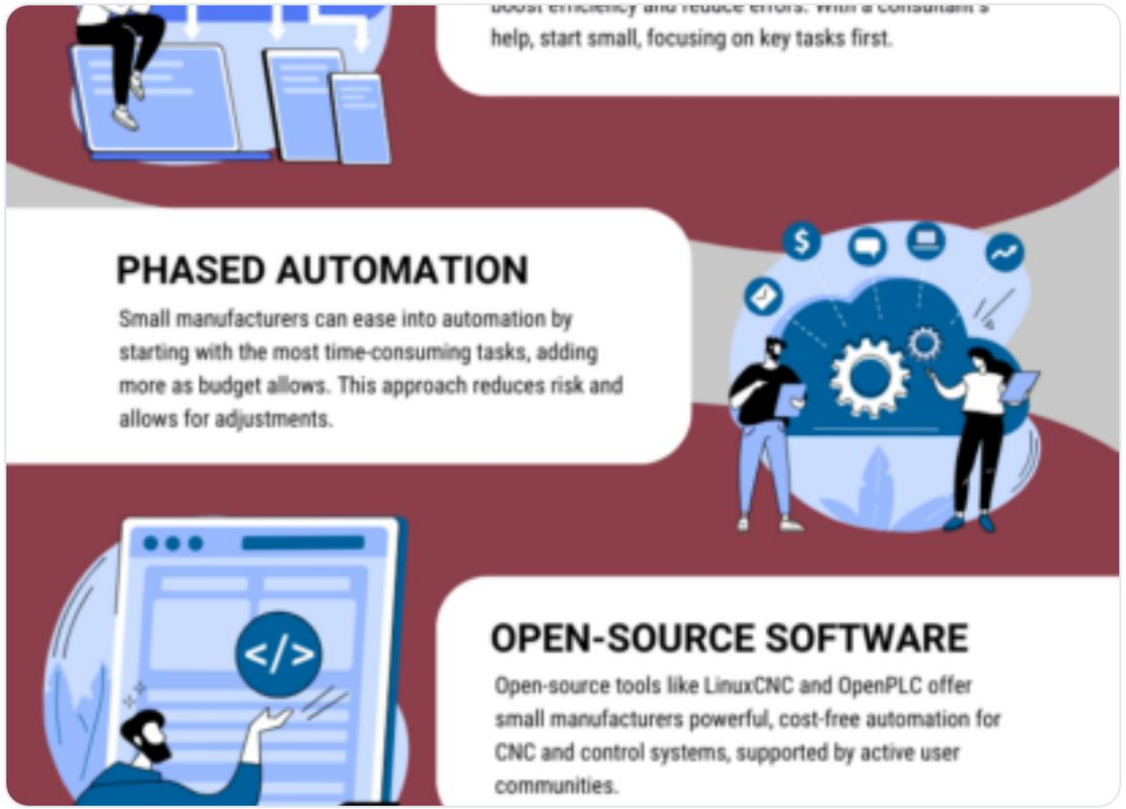
2. Flexibility

- More adaptable than standard robots.
- Easily programmed for new activities.
- Lightweight design.
- Relocatable between work areas as needed.



3. Low Cost

- Generally less expensive than standard robots.
- Requires less maintenance.
- Excellent budget-friendly automation.
- Suitable for companies of any size.



4. Increased Productivity

- Automates repetitive and risky operations.
- Medical device assembly made easy.
- Faster and more accurate than humans.
- Continuous operation (no breaks required).

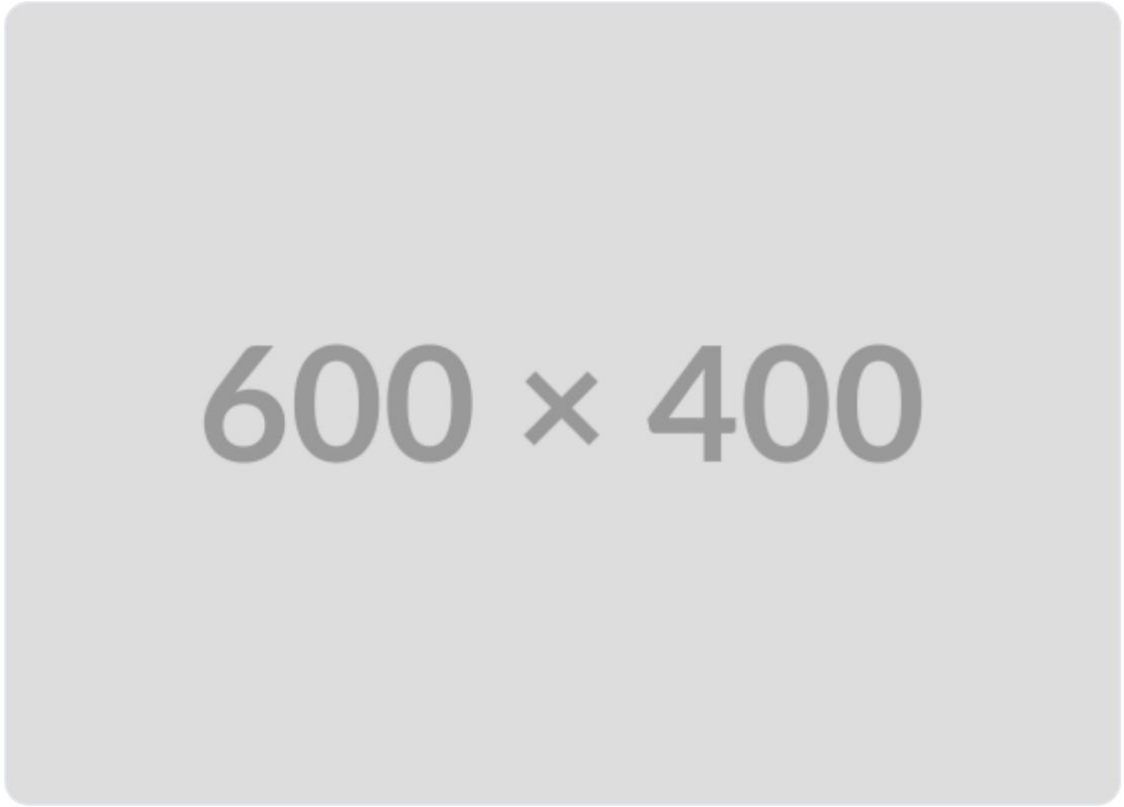


Image Sources



https://static.canadianmetalworking.com/a/human-robot-collaboration-1579878686.jpg?size=1000x

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Source: pngtree.com

Robots vs. Cobots

Understanding the fundamental differences in purpose, design, and operation.

Comparison Overview

Feature	Robot	Cobot
Purpose	Dangerous/Repetitive tasks	Collaborative tasks
Design	Large, heavy, enclosed	Smaller, light, open
Speed/Strength	Very high	Lower, safety-focused

1. Purpose

Robot: Designed to automate tasks that are inherently dangerous, repetitive, or tedious to humans.

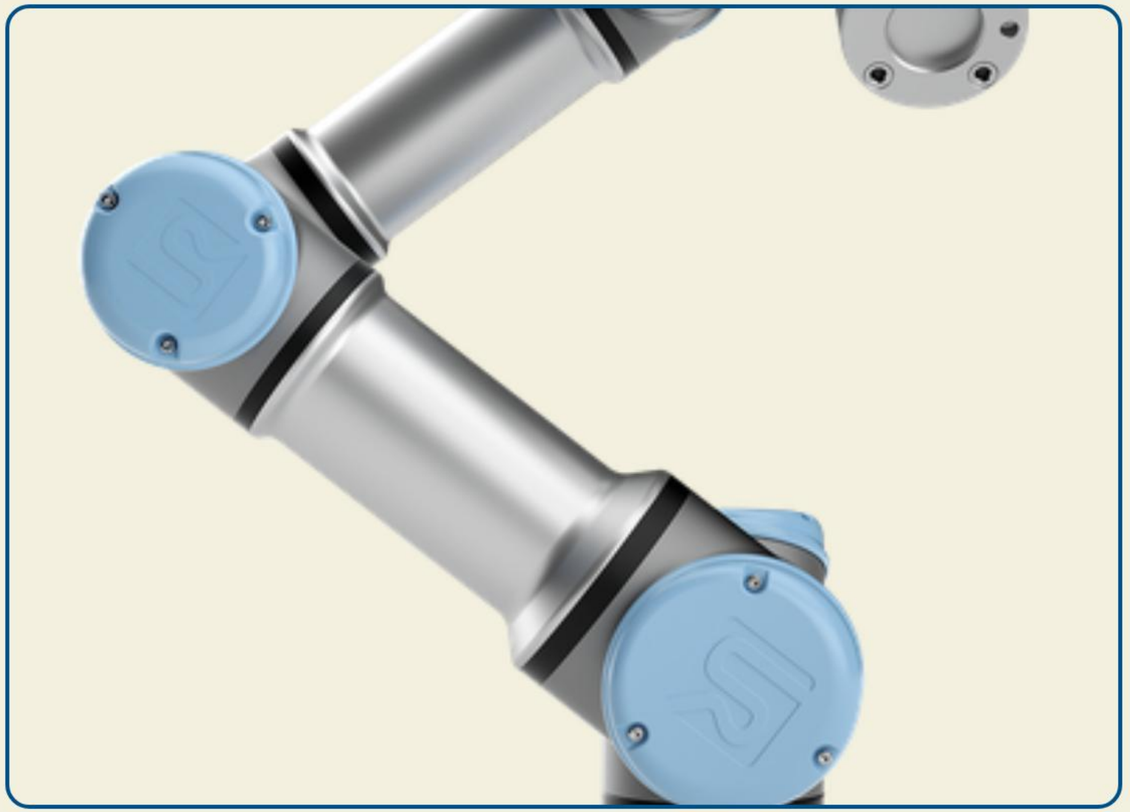
Cobot: Engineered for direct interaction, designed to work side-by-side with humans in a safe, collaborative environment.



2. Design

Robot: Typically large, heavy, and strictly enclosed in safety cages to prevent human interaction.

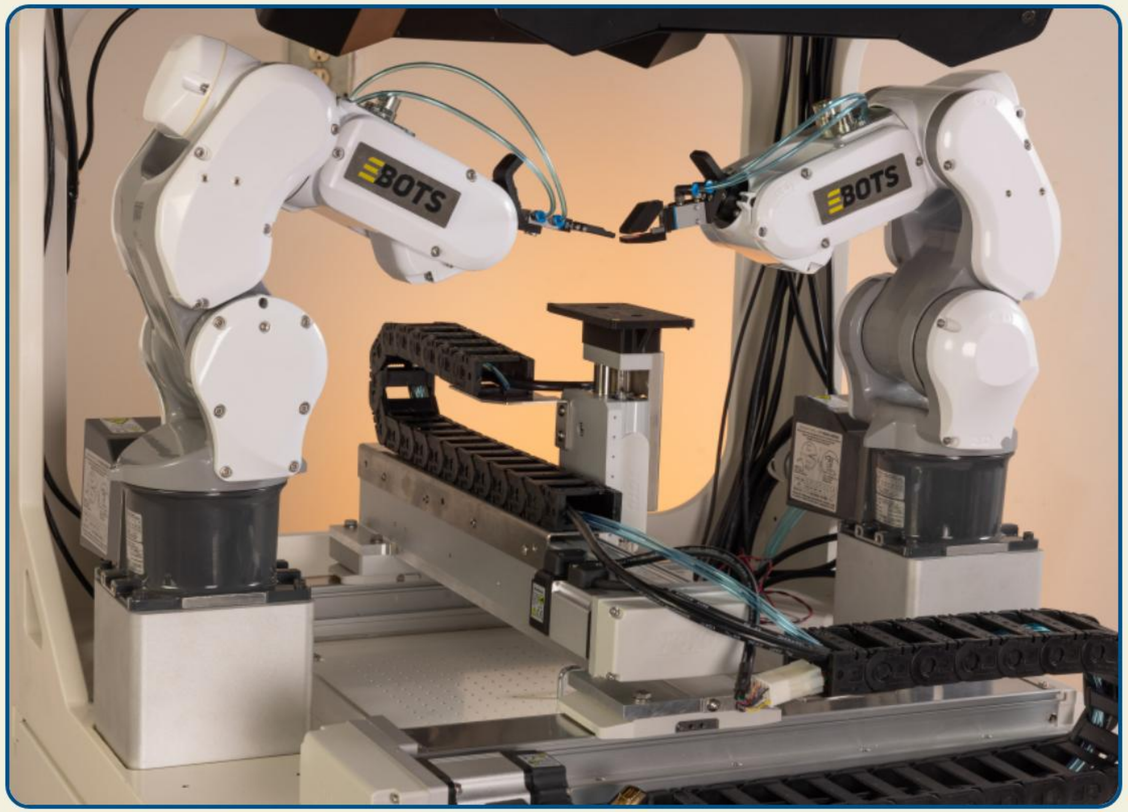
Cobot: Smaller, lightweight, and designed with open frameworks to facilitate proximity to humans.



3. Speed

Robot: High operational speed, suitable for rapid, high-throughput manufacturing.

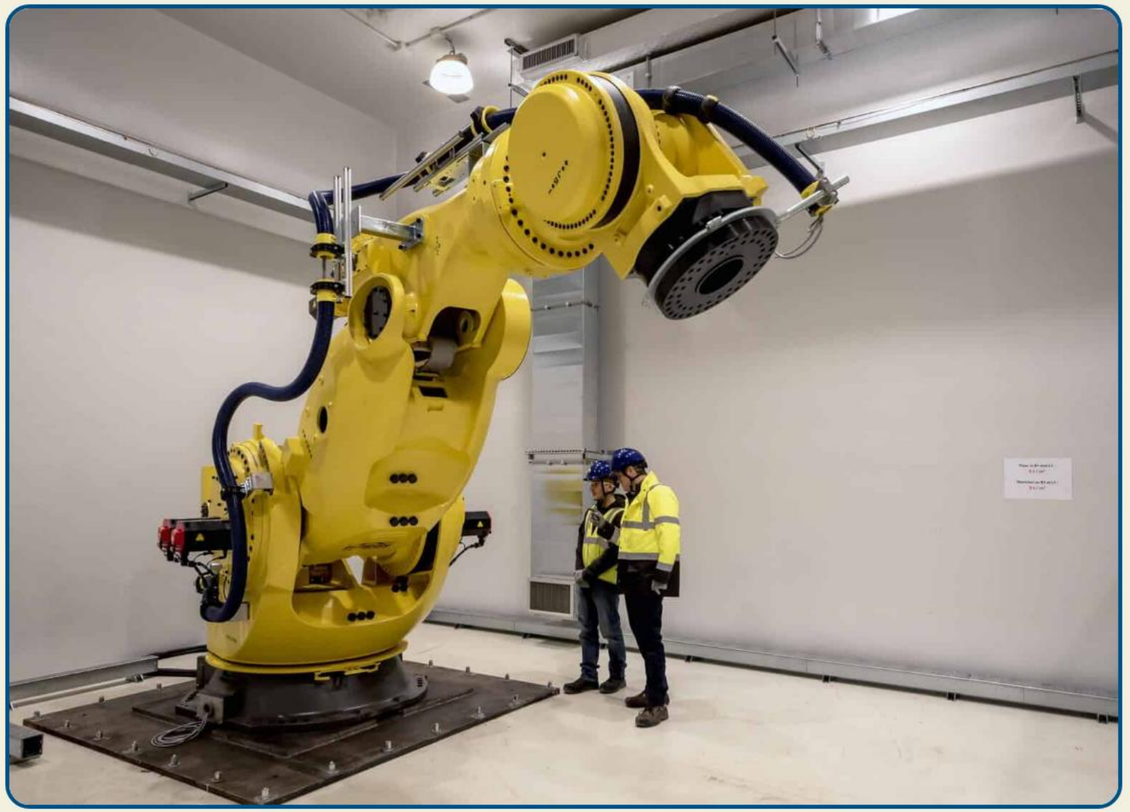
Cobot: Slower operational speeds, prioritized for safety when collaborating with human workers.



4. Strength

Robot: Often capable of exerting extreme force; ideal for heavy industrial lifting and assembly.

Cobot: Generally less strong, focusing on precision, dexterity, and lightweight support tasks.



5. Cost

Robot: Generally more expensive due to specialized safety requirements, high strength, and power.

Cobot: More cost-effective, offering flexible automation that is more accessible to smaller enterprises.

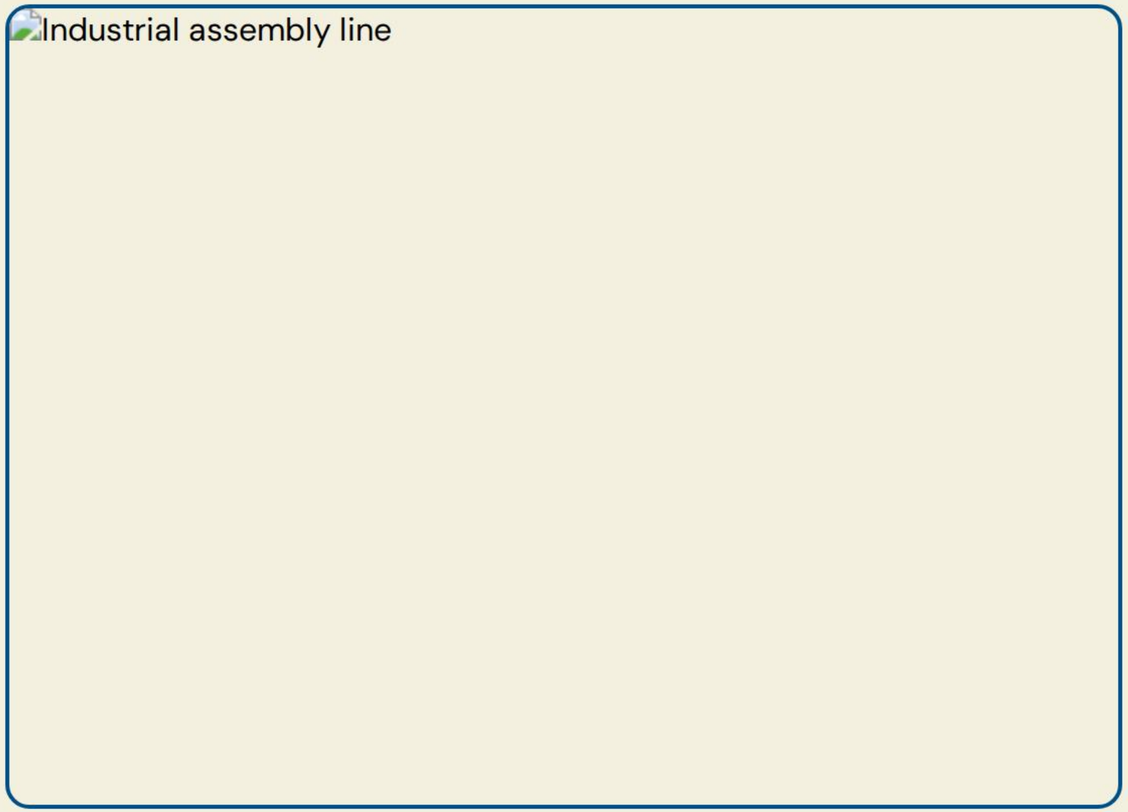


Image Sources



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TYPES OF GEARS





Gears - It's Classification

- Spur Gear
- Helical Gear
- Herringbone Gear
- Bevel Gear
- Worm Gear
- Rack and Pinion
- Internal and External Gear
- Face Gear
- Sprcokets

















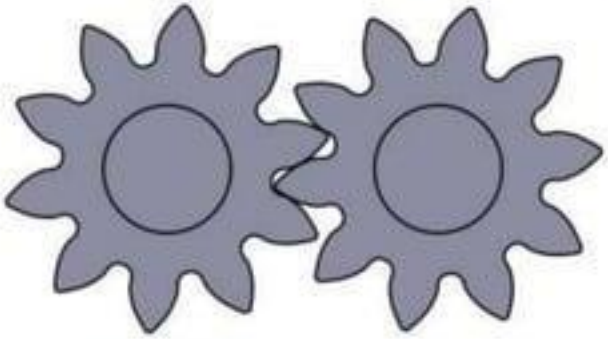


Spur Gear





- Parallel and co-planer shafts connected by gears are called spur gears.
- Spur gears have straight teeth and are parallel to the axis of the wheel.
- The advantages of spur gears are their simplicity in design, economy of manufacture and maintenance, and absence of end thrust.
- Spur gears are known as slow speed gears.
- Spur gears are the most common type of gears.

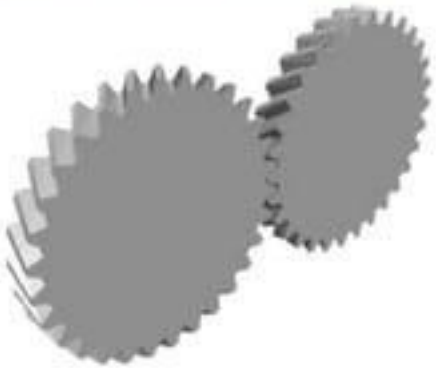




Helical Gear



- Helical gears have their teeth inclined to the axis of the shafts in the form of a helix.
- Helical gears can take higher loads than similarly sized spur gears. The motion of helical gears is smoother and quieter than the motion of spur gears.
- The angle of the helix on both the gear must be same in magnitude but opposite in direction, i.e., a right hand pinion meshes with a left hand gear.
- These gears are usually considered as high speed gears.





Herringbone Gear



- Herringbone gears resemble two helical gears that have been placed side by side.
- A disadvantage of the herringbone gear is that it cannot be cut by simple gear hobbing machines.
- Herringbone is usually referred to as Double helical gear but the fact is that Herringbone is type of double gear with two helical gears joined together correctly of opposite helix and they don't have any clearance between them, whereas in double helical gear there is a groove between the two helical gear given for tool run out for manufacturing.





Bevel/Miter Gear



Intersecting but coplanar shafts connected by gears are called bevel gears. Straight bevel gears can be used on shafts at any angle, but right angle is the most common.

Spiral Bevel Gears

- In these Spiral Bevel gears, the teeth are oblique. Spiral Bevel gears are quieter and can take up more load as compared to straight bevel gears.



Zero Bevel Gears

- Zero Bevel gears are similar to straight bevel gears, but their teeth are curved lengthwise. These curved teeth of zero bevel gears are arranged in a manner that the effective spiral angle is zero.



Worm Gear

- Worm gears are used to transmit power at 90° and where high reductions are required.
- In worm gears, one gear has screw threads. Due to this, worm gears are quiet, vibration free and give a smooth output.
- Worm gears and worm gear shafts are almost invariably at right angles.





Rack and Pinion

- A rack is a toothed bar or rod that can be thought of as a sector gear with an infinitely large radius of curvature. Pinion can be considered as a spur gear.
- Torque can be converted to linear force by meshing a rack with a pinion: the pinion turns; the rack moves in a straight line.
- Such a mechanism is used in automobiles to convert the rotation of the steeing wheel into the left-to-right motion of the tie rod(s).





Internal and External Gear

Internal Gear

- An internal gear is one with the teeth formed on the inner surface of a cylinder or cone.



External Gear

- An external gear is one with the teeth formed on the outer surface of a cylinder or cone.



Face Gear

- Face gears transmit power at (usually) right angles in a circular motion.
- Face gears are not very common in industrial application.



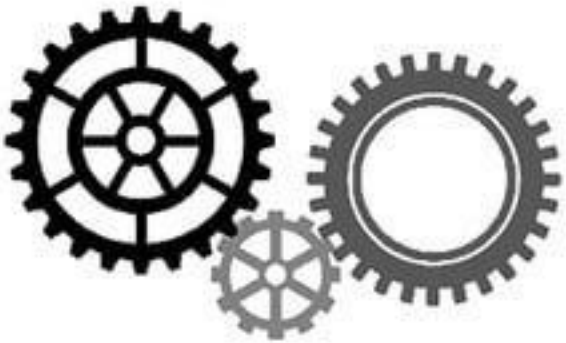


Sprockets



- Sprockets are used to run chains or belts. They are typically used in conveyor systems.
- They are used in chain drives, in motorcycles and bicycles.







Classification according to axis of shaft

- Parallel
  - Spur Gear
  - Helical Gear
  - Rack and Pinion
- Intersecting
  - Bevel Gear
- Non-intersecting and Non-parallel
  - Worm and worm gears

THANK YOU



- Mrunal Mohadikar (38)